

Design and Development of a Low Cost Potentiostat for Differential Pulse Voltammetry

by

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Abstract

Pharmaceutical residues, including antiretrovirals, are increasingly found in surface and waste water, but measuring them requires expensive laboratory instruments. This project presents the OpenStat, an open-source, battery-powered potentiostat for differential pulse voltammetry (DPV), aimed at detecting efavirenz and oseltamivir in water. It combines an AD5941 analog front-end with an ESP32-S3 microcontroller and is operated through a display, buttons and a rotary encoder, or wirelessly through a Wi-Fi web interface. Custom firmware controls the DPV pulse timing and auto-ranges the current across eight transimpedance gains. The waveform is accurate to within one LSB, and after calibration all eight current ranges read within 0.44% of a reference multimeter. Dummy-cell responses were within 1.9% of expected. DPV scans of a 0.1 M salt blank and 5 mM potassium ferricyanide on gold screen-printed electrodes matched a commercial PalmSens4 within its scan-to-scan repeatability, apart from a 24 mV potential offset.

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Nomenclature

List of Symbols

Symbol	Description	Unit
Electrochemical cell		
R_s	Solution resistance	Ω
R_u	Uncompensated resistance between the RE and WE	Ω
C_{dl}	Double-layer capacitance	F
R_{ct}	Charge-transfer resistance	Ω
Z_W	Warburg (diffusion) impedance	Ω
I_c	Capacitive (charging) current	A
I_f	Faradaic current	A
n	Number of electrons transferred	–
F	Faraday constant	$\mathrm{C\,mol^{-1}}$
A	Electrode area	$\mathrm{cm^2}$
D	Diffusion coefficient	$\mathrm{cm^2\,s^{-1}}$
C^*	Bulk analyte concentration	$\mathrm{mol\,cm^{-3}}$
t	Time	s
Transimpedance amplifier and stability		
R_{TIA}	Transimpedance (gain) resistor	Ω
C_f	Feedback capacitor across R_{TIA}	F
R_L	Internal HSTIA load resistor	Ω
R_Σ	Total series resistance to the solution, $R_u + R_L$	Ω
Z_f, Z_c	Feedback and cell impedance	Ω
$A(s)$	Open-loop gain of the amplifier	–
$NG(s)$	Noise gain	–
GBW	Gain-bandwidth product	Hz
f_z	Noise-gain zero frequency	Hz
f_p	Noise-gain pole frequency	Hz
f_c	Loop-gain crossover frequency	Hz
ϕ_{PM}	Phase margin	°
V_{BIAS}	LPDAC bias voltage (sets the RE potential)	V
V_{ZERO}	LPDAC zero voltage (sets the WE potential)	V

Symbol	Description	Unit
V_{TIA}	TIA output voltage	V
D_{12}	12-bit LPDAC code	–
DPV waveform and measurement		
E	Cell potential, $V_{\mathrm{ZERO}} - V_{\mathrm{BIAS}}$	V
E_{start}, E_{end}	Start and end potential of the scan	V
ΔE	Potential step	V
ΔE_{s}	Step potential	V
ΔE_{p}	Pulse amplitude	V
t_{p}	Pulse width	s
t_{b}	Base dwell, $T - t_{\mathrm{p}}$	s
T	Pulse period	s
N	Number of steps in a scan	–
T_{s}	ADC (sinc2) output period	s
N_{w}	Number of samples averaged per window	–
T_{w}	Averaging window length, $N_{\mathrm{w}}T_{\mathrm{s}}$	s
ΔI	Differential current, $I_{\mathrm{end}} - I_{\mathrm{before}}$	A
I_{raw}	Uncalibrated current	A
G_k	Calibration gain for current range k	–
O_k	Calibration offset for current range k	A
Hardware		
L_{bead}	Ferrite bead inductance	H
R_{DC}	Ferrite bead DC resistance	Ω
f_0	Filter resonant frequency	Hz
Q	Filter quality factor	–
T_{rise}	Allowed trace temperature rise	°C
W	Trace width	mil

Acronyms and Abbreviations

ADC	Analog-to-Digital Converter
AFE	Analog Front-End
BLE	Bluetooth Low Energy
BOM	Bill of Materials
CA	Chronoamperometry
CE	Counter Electrode
CV	Cyclic Voltammetry
DAC	Digital-to-Analog Converter
DC	Direct current
DMM	Digital Multimeter
DPV	Differential Pulse Voltammetry
EIS	Electrochemical Impedance Spectroscopy
EPR	Extended Producer Responsibility
ESL	Equivalent Series Inductance
GBW	Gain-Bandwidth Product
GUI	Graphical User Interface
HSTIA	High-Speed Transimpedance Amplifier
IC	Integrated Circuit
IoT	Internet of Things
LCMS	Liquid Chromatography-Mass Spectrometry
LDO	Low Dropout Regulator
LOD	Limit of Detection
LPDAC	Low-Power Digital-to-Analog Converter
LPTIA	Low-Power Transimpedance Amplifier
LSB	Least Significant Bit
LSV	Linear Sweep Voltammetry
MCU	Microcontroller Unit
NVS	Non-Volatile Storage

Op-amp	Operational amplifier
PCB	Printed Circuit Board
PGA	Programmable Gain Amplifier
PS4	PalmSens4
PSRAM	Pseudo-Static Random-Access Memory
PSRR	Power Supply Rejection Ratio
RE	Reference Electrode
SNR	Signal-to-Noise Ratio
SoC	System-on-Chip
SPE	Screen-Printed Electrode
SPI	Serial Peripheral Interface
SRF	Self-Resonant Frequency
SWV	Square Wave Voltammetry
TIA	Transimpedance Amplifier
UWED	University of Washington Electrochemical Device
WE	Working Electrode

Definitions

A

Analyte A specific substance or chemical component that is the subject of analysis.

B

Base Dwell The time the potential sits at the base level before a pulse is applied, in each step.

C

Code The integer the firmware writes to the DAC to set its output voltage.

Compliance Voltage The maximum voltage a potentiostat can apply between the counter electrode and the working electrode in order to keep the working electrode at the desired potential.

E

Electrochemically Reversible System A system where the electron transfer kinetics between the electrode and the analyte are so fast that they are practically instantaneous compared to the speed of the measurement taken.

V

Voltammetric Anything relating to, or measured by, voltammetry.

Voltammetry A family of electrochemical techniques where the potential of an electrode is controlled and the current that flows as a result is measured.

Voltammogram A plot of current versus applied potential for a given electrochemical reaction.

Chapter 1

Introduction

1.1 Background and Motivation

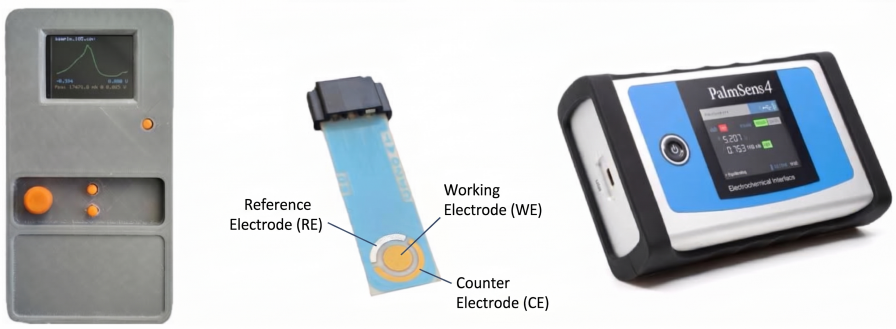
In South Africa, a critical public health and environmental challenge is the accumulation of active pharmaceutical ingredients, specifically antiretroviral and other antiviral medication, in local water systems. The amount of pharmaceutical compounds has been measured at concentrations up to 102 µg/L in surface water [1, 2]. Monitoring these specific pharmaceutical compounds at trace levels requires highly sensitive analytical techniques. The established method, liquid chromatography–mass spectrometry (LCMS), is laboratory-based and costly. South Africa currently has no monitoring programme or regulatory limit for these compounds [1]. Differential Pulse Voltammetry (DPV) was selected as a technique to detect these pharmaceutical compounds in the water due to its high sensitivity [3]. DPV methods have been reported for two such compounds, the antiretroviral efavirenz and the influenza antiviral oseltamivir [4, 5], which are the target compounds of this project. DPV superimposes small pulses of potential onto a staircase potential ramp. By measuring the current difference before and near the end of each pulse, the technique reduces background charging current and achieves high sensitivity [6].

1.2 Problem Statement

The primary barrier to testing water in South Africa is that the tools required to perform accurate voltammetric analysis are costly and lack portability [7, 8]. Commercially available potentiostats, such as the PalmSens4 seen in Figure 1.1, serve as the gold standard for electrochemical analysis. However, these instruments are cost-prohibitive and operate as closed-source devices [9]. While various open-source alternatives have been developed over the years, many remain tethered to external computers [7] for power and data processing, or rely on proprietary applications that can become obsolete. Web-hosted designs remove this dependence but have no interface of their own, and none of the designs surveyed in Chapter 2 combines an integrated electrochemical front end with both an onboard display and a browser interface. This presents a distinct need for an autonomous, wireless and

low-cost potentiostat capable of conducting sensitive DPV measurements independently in the field [10].

Figure 1.1: The OpenStat developed in this project (Left), Metrohm C220 screen-printed electrode (SPE) (Centre), and PalmSens4 reference potentiostat (Right, Source: [11]).



1.3 Project Aim and Objectives

The primary aim of this project is to design, build and validate a low-cost, wireless potentiostat for the DPV detection of efavirenz and oseltamivir in water, as a step towards screening in the field.

To meet this aim, the device must satisfy the following objectives, each of which is verified in Chapter 6:

1. Apply the DPV waveform specified for the target compounds (Table 4.1) with a potential resolution of 1 mV or better, pulses of 25 mV ± 1 mV and a step period within ±1 %.
2. Measure currents up to ±750 µA with sub-nanoampere resolution on the most sensitive range and a gain error of at most 1 % on every range after calibration.
3. Agree with the ideal response of dummy cells, and with a PalmSens4 on standard solutions within that instrument's own scan-to-scan repeatability.
4. Operate standalone through an onboard display and controls, and remotely from a web browser over Wi-Fi.
5. Run from an internal battery for at least one eight-hour working day.
6. Keep the cost of the prototype below one tenth of the cost of a commercial portable potentiostat, the PalmSens4, listed at approximately R55 400 [12].

Objectives 1 to 3 ensure that the measurements can be trusted against a laboratory instrument, objectives 4 and 5 allow the device to be used in the field without a computer

or mains power, and objective 6 addresses the cost barrier identified in Section 1.2.

The approach taken, justified in Chapters 2 to 5, combines an AD5941 analog front end (AFE), specified at the start of the project, with an ESP32-S3 microcontroller with native Wi-Fi, an onboard display, two push buttons and a rotary encoder, a three-electrode interface for screen-printed electrodes, and C++ firmware that runs the measurement, the local interface and a web server.

1.4 Scope and Contributions

The scope of this project focuses on the design, development and analytical validation of the potentiostat hardware and firmware. To validate the DPV measurement performance, the scope was restricted to laboratory-prepared standard solutions of salt and ferricyanide in a supporting electrolyte, measured on both the OpenStat and a PalmSens4. The detection of efavirenz and oseltamivir in real-world water samples is outside the scope of this project, as it would require additional sample preparation, extraction and purification steps to isolate the target analytes and the addition of a supporting electrolyte [13]. Standard solutions of efavirenz and oseltamivir were also not measured, because access to these compounds was limited; ferricyanide was used instead as a well-characterised redox probe.

The main contributions of this project are an AD5941-based potentiostat that can be operated both from its own display and from a web browser, firmware-timed DPV for sensitive measurements and a stepwise validation of the device against ideal loads and a PalmSens4 (Chapter 6).

1.5 Report Overview

Chapter 2 reviews the electrochemistry of DPV, the circuit principles of a potentiostat and existing commercial and open-source designs, and identifies the gap this project addresses. Chapter 3 models the measurement signal chain and analyses the choice and stability of the current amplifier. Chapter 4 sets out the requirements and describes the hardware that meets them. Chapter 5 describes the firmware, focusing on pulse timing, current sampling and range selection. Chapter 6 tests the device in five steps, from the excitation waveform alone to a redox probe on a real electrode, and compares it with the PalmSens4. Chapter 7 assesses the environmental impact and cost of the device, and Chapter 8 concludes and recommends future work.

Chapter 2

Literature Review

This chapter reviews the literature relevant to the design of the OpenStat for DPV. The purpose of this review is to establish the electrochemical and hardware fundamentals required to design such an instrument, and to survey the existing landscape of commercial and open-source potentiostat designs.

2.1 Electrochemical Sensing

A potentiostat controls the potential between the reference and working electrode while simultaneously measuring the current flowing between the counter and the working electrode. This is a popular three-electrode setup for electrochemical measurement, illustrated in Figure 2.1. The Working Electrode (WE) is the focal area of electrochemical reactions [14]. In a three-electrode system the potential around it is measured and manipulated to study the desired catalyst or analyte.

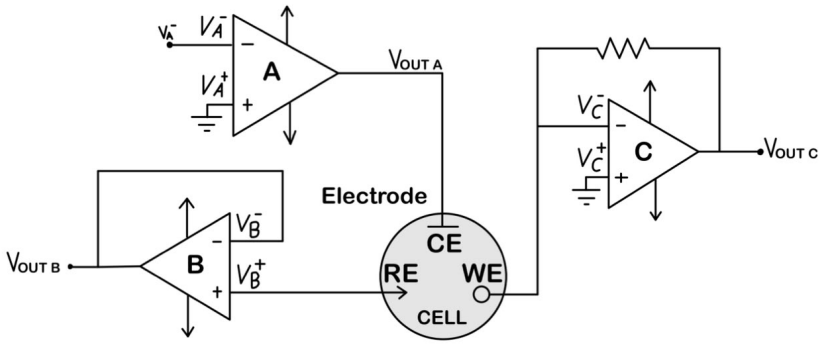


Figure 2.1: The classic three-op-amp potentiostat topology, showing the control amplifier (A), RE buffer (B) and TIA (C) connected to the three electrodes.

The Reference Electrode (RE) provides a stable, known potential that is used as a benchmark to control and monitor the working electrode's potential [15]. Current flow through the reference electrode is zero to ensure that its potential remains accurate [15].

The Counter Electrode (CE) (also called the auxiliary electrode) completes the circuit, allowing current to be sourced or sunk between it and the working electrode, which prevents the reference electrode from being affected [15]. The reaction at the counter electrode should be monitored, as corrosion at the electrode could interfere with measurement accuracy [16]. The three-electrode setup is the foundational hardware configuration for the voltammetric techniques discussed in Section 2.1.1, namely Cyclic Voltammetry (CV), Linear Sweep Voltammetry (LSV), Square Wave Voltammetry (SWV), and Differential Pulse Voltammetry (DPV).

2.1.1 Voltammetry as a Family of Techniques

In any voltammetric measurement, applying a voltage to an electrode generates two types of current:

- **Faradaic current:** The actual current generated by the transfer of electrons during the oxidation or reduction of the target analyte [17]. This is the desired signal.
- **Charging (capacitive) current:** The background current generated simply by charging the electrical double layer at the electrode surface, which behaves like a capacitor [17]. This produces an unwanted background current. The double layer is modelled as the capacitance C_{dl} in Section 2.1.2.

The same three-electrode setup handles each of the following techniques in a different way:

- **Linear Sweep (LSV) and Cyclic Voltammetry (CV):** The potentiostat applies a continuous, linear voltage ramp (or a triangular sweep for CV). The counter electrode continuously sinks or sources the resulting Faradaic current.
- **Differential Pulse (DPV) and Square Wave Voltammetry (SWV):** The potentiostat applies discrete voltage pulses or steps and samples the current at set times, so that the charging current can be subtracted (Section 2.1.3). This makes DPV and SWV the preferred techniques for trace-level detection.

2.1.2 Equivalent Circuit of the Cell

The two currents described in Section 2.1.1 can be represented by an electrical model of the electrode–electrolyte interface. The widely used Randles equivalent circuit [18], shown in Figure 2.2, models this interface as the solution resistance R_s in series with the double-layer capacitance C_{dl}, which is in parallel with the charge-transfer resistance R_{ct} and the Warburg impedance Z_W. The charging current flows into C_{dl}, while the Faradaic current flows through R_{ct} and Z_W. R_{ct} represents the resistance to electron transfer at the electrode surface, and Z_W accounts for the diffusion of analyte towards the electrode

at low frequencies [19, 20]. The time constant $R_s C_{dl}$ sets how quickly the double layer charges after each potential step, which is central to how DPV rejects the charging current.

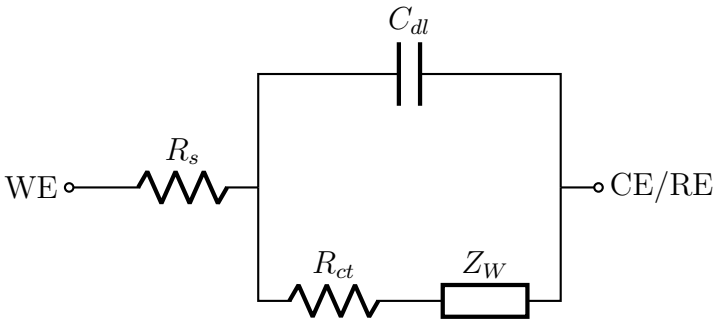


Figure 2.2: Randles equivalent circuit modelling the electrochemical cell interface.

2.1.3 Differential Pulse Voltammetry

DPV was chosen as the focus of this project for its high sensitivity, ability to reject capacitive charging current and strong fit for the intended application [3]. Its excitation waveform is a staircase ramp with a small potential pulse superimposed on each step, as shown in Figure 2.3. The waveform is defined by the step potential, the pulse amplitude and the pulse width [21].

Current is not measured continuously. Two discrete samples are taken per step, I_{before} immediately before the pulse is applied, and I_{end} at the end of the pulse (labelled i_1 and i_2 in Figure 2.3). Their difference,

$$\Delta I = I_{\text{end}} - I_{\text{before}}, \tag{2.1}$$

is plotted against the step potential to form the voltammogram.

The sensitivity of DPV comes from when these samples are taken. After a potential step ΔE, the capacitive current decays exponentially [22],

$$I_c(t) = \frac{\Delta E}{R_s}\, e^{-t/R_s C_{dl}}, \tag{2.2}$$

while the Faradaic current follows the Cottrell equation [19],

$$I_f(t) = \frac{nFA\sqrt{D}\,C^*}{\sqrt{\pi t}}, \tag{2.3}$$

where n is the number of electrons transferred, F is Faraday's constant, A is the electrode area, D is the diffusion coefficient and C^* is the bulk analyte concentration. Because the exponential decays far faster than $1/\sqrt{t}$ [19, Sec. 1.2.4, Sec. 5.2.1], sampling late in the

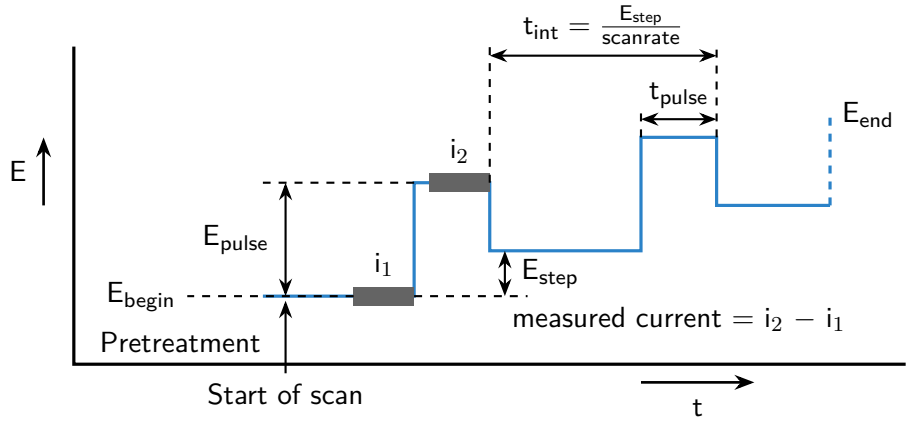
step and late in the pulse allows the capacitive current to decay almost completely, so that $I_{c,\text{before}} \approx I_{c,\text{end}} \approx 0$ and

$$\Delta I \approx I_{f,\text{end}} - I_{f,\text{before}}. \tag{2.4}$$

The full expansion is given in Appendix D.1. In practice this sets a minimum pulse width, since sampling at $t = 5R_sC_{dl}$ leaves less than 1 % of the initial capacitive current ($e^{-5} \approx 0.007$). Because ΔI approximates the derivative of the Faradaic current–potential curve, the voltammogram forms a peak which, for a reversible system, lies half a pulse amplitude away from the half-wave potential $E_{1/2}$ [19].

In continuous sweeps such as CV and LSV, by contrast, charging current is present throughout the measurement. The DPV signal is therefore dominated by Faradaic current, which gives a much higher Signal-to-Noise Ratio (SNR) and lower limits of detection. The cost is reduced time resolution [17].

Figure 2.3: Waveform profile for DPV. Grey blocks represent the current sampling windows. Adapted from [23].



2.1.4 Application to Antiviral Drug Residues in Water

South Africa's large antiretroviral treatment programme, combined with incomplete removal at wastewater treatment plants, releases these drugs into surface water [1]. Efavirenz in particular has been measured at concentrations up to 102 µg/L in surface water near treatment plant discharges [24], yet South Africa has no monitoring programme or regulatory limit for these compounds [1]. Existing measurements rely on laboratory chromatography–mass spectrometry (LCMS), which is costly and unsuited to field use.

Target analytes in these applications may be electroactive. These can include antiretrovirals such as efavirenz, which undergoes irreversible oxidation and has been quantified by DPV on disposable pencil graphite electrodes [4]. Oseltamivir, an influenza antiviral, has been quantified by DPV at modified paste microelectrodes [5] and shows a voltammetric

response at gold electrodes [25]. These studies used benchtop instruments and, in most cases, modified electrodes or adsorptive pre-concentration to reach low detection limits.

To set the sensitivity required of the instrument, Equation 2.3 can be evaluated for the target compounds. For efavirenz ($M \approx 316\,\mathrm{g\,mol^{-1}}$) at $1\,\mu\mathrm{g\,L^{-1}}$ ($\approx 3\,\mathrm{nmol\,L^{-1}}$) on a 4 mm gold electrode ($A \approx 0.126\,\mathrm{cm^2}$), assuming one electron is transferred and $D \approx 5 \times 10^{-6}\,\mathrm{cm^2\,s^{-1}}$, the Faradaic current 50 ms after a potential step is about 0.2 nA. The DPV peak is only a fraction of this current, and both scale linearly with concentration, from about 2 nA at $10\,\mu\mathrm{g\,L^{-1}}$ down to picoamperes at nanograms per litre. The instrument therefore needs sub-nanoampere resolution (Objective 2 in Section 1.3), while the lowest concentrations still require preconcentration or a modified electrode, as in [4, 5].

2.2 Potentiostat Fundamentals

This section explores the fundamentals of a potentiostat, namely its role, design, performance parameters, and implementation approaches.

2.2.1 The Role of a Potentiostat

A potentiostat consists of three functional blocks: a control amplifier, a reference-electrode buffer and a TIA (Figure 2.1). For reference electrode buffering, the electrode must remain perfectly stable at a known half-cell potential. To prevent the voltage from drifting, zero current is allowed to flow through the electrode. This is achieved using an electrometer, an operational amplifier with an extremely high input impedance [26].

The control amplifier drives the system by constantly comparing the desired input voltage, set by a DAC, with the actual voltage measured at the reference electrode. The resulting error signal is minimised by forcing current through the counter electrode and into the solution, which actively compensates for any changes in the electrochemical cell's resistance. The TIA holds the working electrode at a 0 V reference while catching every electron produced or consumed by the reaction [19]. These electrons generate a small Faradaic current that is converted into a potential, which can then be measured by an Analog-to-Digital Converter (ADC) [19]. The precision of the measurements made by the TIA depends not only on the AFE's noise performance but also on when the sample is taken. As discussed in Section 2.1.3, the capacitive and Faradaic currents are distinguished by their differing decay times. Therefore, to sample accurately, precision timing and specific, predictable sampling windows are required. The occurrence of a timing jitter would directly degrade the signal-to-noise ratio regardless of the AFE's noise performance. To ensure accurate measurements key performance parameters need to be considered when building and designing a potentiostat.

2.2.2 Key Performance Parameters

This section establishes the criteria used to evaluate the hardware, namely current measurement range and resolution, compliance voltage, noise floor, bandwidth, and technique support. Chapter 6 evaluates the OpenStat against the current range and resolution, noise floor and technique support.

- **Current measurement range and resolution:** In drug residue sensing, very small Faradaic signals are generated. The potentiostat must be able to accurately measure down to the nanoampere (nA) or picoampere (pA) range without clipping or saturating the signal.
- **Compliance voltage:** The control amplifier must drive the cell resistance $R_s + R_{ct}$ of Section 2.1.2. If this resistance is high, the control amplifier needs a high compliance voltage to push the necessary current through the counter electrode [27].
- **Noise floor and Limit of Detection (LOD):** The LOD is constrained by the noise floor of the potentiostat's AFE. In high-sensitivity applications, the TIA requires large feedback resistors to convert picoampere-level Faradaic currents into measurable voltages. However, these resistive components inherently generate thermal noise. Therefore, a device targeting nanoampere or picoampere signals requires a potentiostat topology with an exceptionally low electrical noise floor [28, 29].
- **Bandwidth / scan rate limits:** This is the maximum speed at which the potentiostat can alter the applied potential. Fast techniques such as SWV require high-bandwidth operational amplifiers to avoid signal distortion [19].
- **Technique support:** A system's ability to run CV, DPV, or Electrochemical Impedance Spectroscopy (EIS) is set by its Digital-to-Analog Converter (DAC) and any on-chip signal generator [30, 31].

2.2.3 Analog Front-End Approaches

Transitioning from the theoretical 3-op-amp topology to a physical implementation requires selecting an appropriate hardware architecture. To combat bulky benchtop potentiostats, two approaches to AFEs were developed. First, a discrete circuit using individual operational amplifiers. Second, utilising a fully integrated System-on-Chip (SoC) AFE, such as the AD5941. Each approach presents distinct engineering trade-offs which are summarised in Table 2.1.

Table 2.1: Comparison of discrete op-amp and integrated AFE IC approaches.

Approach	Advantages	Trade-offs
Discrete op-amp design	Architectural flexibility, capable of massive compliance voltages, component-level tuning for extreme precision.	Large physical PCB footprint, high power consumption, higher cost of materials.
Integrated AFE ICs	Millimetre-scale physical footprint, very low power consumption, built-in hardware digital filtering.	Limited compliance voltage ($\leq 3\,\mathrm{V}$), fixed internal current measurement ranges, limited deep customisation, requires complex firmware development.

2.3 Survey of Existing Potentiostat Designs

The democratisation of electrochemical instrumentation has led to a rapid expansion of custom potentiostat designs. This section surveys the architectural evolution of these devices, and the comparison between them is summarised in Table 2.2.

Table 2.2: Comparison of surveyed potentiostat designs.

Design / Platform	MCU / AFE Architecture	Techniques Supported	Connectivity & Interface	Onboard Display?	Open-Source Completeness
Commercial (e.g. PS4 [11])	Proprietary	CV, DPV, SWV, EIS, CA	USB / Proprietary GUI	Yes	Closed source
CheapStat [8]	Atmel AVR / Discrete	CV, SWV, CA	None / Basic LCD output	Yes	Full (PCB/Firmware)
DStat [9]	Atmel XMEGA / Discrete	CV, DPV, SWV, CA	USB / Python PC GUI	No	Full (PCB/Firmware/GUI)
UWED [7]	BLE MCU / Discrete	CV, DPV, SWV	Bluetooth / Custom App	No	Partial (Hardware focus)
HunStat2 [31]	Mixed / AD5941 SoC	CV, DPV, SWV, EIS	USB / serial data logging	No	Full (Hardware/Firmware)
NanoStat [32]	ESP32 / Discrete	CV, SWV, CA	Wi-Fi / Web GUI	No	Full (Hardware/ GUI)
OpenStat (this work)	ESP32-S3 / AD5941 SoC	DPV	Wi-Fi / Web GUI and onboard GUI	Yes	Full (Hardware/Firmware)

Each design from Table 2.2 is discussed in more detail below, highlighting the evolution of potentiostat architectures and their trade-offs.

2.3.1 Commercial Instruments

Commercial systems such as the PalmSens4 [11], EmStat [33] and Metrohm Autolab [34] are the benchmark for electrochemical analysis, offering wide dynamic ranges (pA to mA), factory calibration and mature software. However, they are expensive, bulky and closed-source, which prevents their integration into low-cost portable systems. Their value lies in providing the performance reference against which open-source and custom hardware is validated [35].

2.3.2 Discrete Open-Source Designs

CheapStat [8] showed that basic analytical electrochemistry was achievable with a Bill of Materials (BOM) under $100. DStat [9] closed much of the performance gap, reaching near-commercial sensitivity with a discrete analog front-end and supporting CV, SWV, DPV and CA through a Python Graphical User Interface (GUI). However, it depends on a USB-connected PC for power and control, which rules out field use. Smartphone-integrated designs such as the UWED [7] achieved portability by moving processing and display to a phone over Bluetooth Low Energy (BLE), but this ties the hardware to a companion app that must be maintained across operating system updates.

2.3.3 AFE-Based and Web-Hosted Designs

Integrated mixed-signal AFEs have largely replaced discrete op-amp circuits in recent designs. Systems such as ABE-Stat [36] and HunStat2 [31] characterise this generation by using SoCs such as the AD5941; they support techniques up to EIS with sub-milliwatt power consumption, low noise and a small footprint [31]. HunStat2 replaced its predecessor HunStat [30], which generated the waveform with the microcontroller's own 10-bit DAC and was limited to DC techniques. However, configuring the AFE requires complex register-level firmware, and these devices typically operate as data-loggers with no user interface, relying on raw serial output for analysis [31]. To remove the dependence on a companion app, ESP32-based designs such as NanoStat [32] and DM-Stat [37] host an embedded web server. Not every ESP32 design does this though. For example, the SaxStat [38] uses the ESP32 only as a USB serial bridge to a desktop application.

These web-hosted designs are headless and cannot be operated without a second device. Early designs such as CheapStat included a basic display, but none of the surveyed systems combines an integrated AFE, a browser interface and a standalone onboard display. Such a dual-interface instrument would allow quick standalone measurements in the field while retaining web-based data access, and this is the gap addressed by this project.

Chapter 3

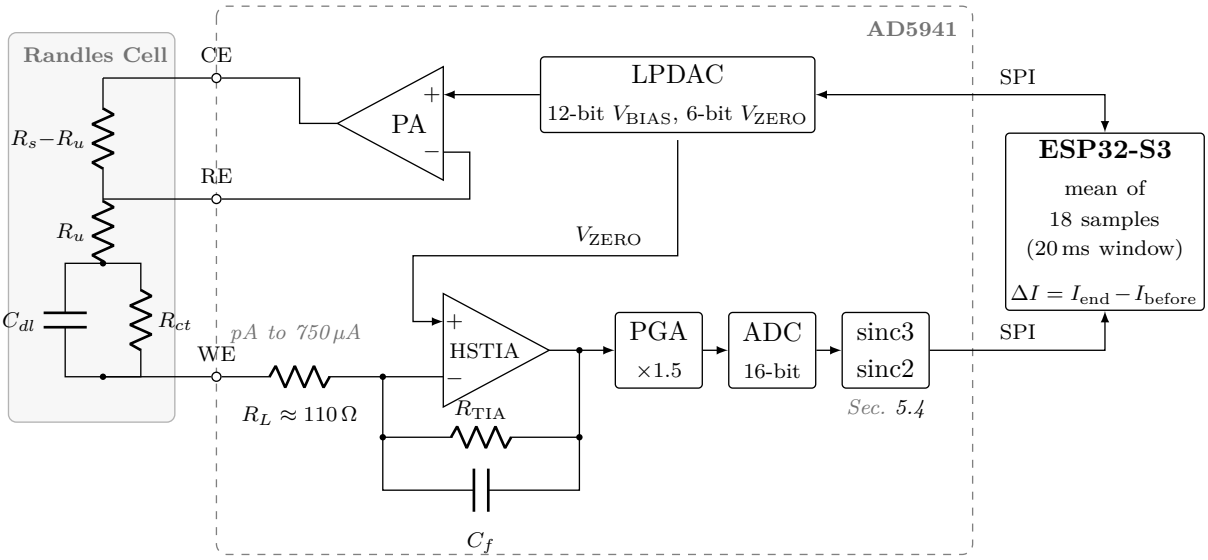
System Modelling and Design

This chapter models the measurement path of the OpenStat and uses the model to choose between the high-speed TIA (HSTIA) and the low-power TIA (LPTIA) of the AD5941. The conditions under which the chosen amplifier remains stable with an electrode connected are also discussed.

3.1 Signal Chain

Figure 3.1 shows the measurement path of the OpenStat. The low-power DAC (LPDAC) of the AD5941 sets V_{BIAS}, and the potentiostat amplifier drives the CE until the RE reaches the same potential. The HSTIA holds the WE at V_{ZERO} and converts the cell current to a voltage, which is amplified, digitised and filtered before the ESP32-S3 averages it into I_{before} and I_{end} (Section 5.4).

Figure 3.1: Measurement signal chain of the OpenStat. The Randles Cell has R_s split at the RE into the part compensated by the control loop, $R_s - R_u$, and the uncompensated resistance R_u between the RE and WE; Z_W is omitted because it only affects the low-frequency response. Figure derived from [39]



In the feedback loop two aspects need to be considered, the fixed internal load resistor $R_L \approx 110\,\Omega$ [39, pp. 10, 50], and the capacitor C_f across the gain resistor R_{TIA}, which is programmable from 1 pF to 63 pF [39, p. 52].

3.2 Choice and Operating Limits of the TIA

The AD5941 has two TIAs that can measure the WE current. The LPTIA is intended for continuous, low-power sensing and offers gain resistors up to 512 kΩ [39, p. 10], which allows smaller currents to be amplified.

3.2.1 High-Speed or Low-Power TIA

According to the datasheet, the recommended configuration for voltammetry is to use the HSTIA. This drives the CE from the LPDAC through the potentiostat loop, disconnects the output capacitors so that the potential can step quickly, and measures the WE current on the HSTIA [39, Table 21, p. 38]. The OpenStat follows this configuration.

The LPTIA was not ruled out on speed; with a small feedback capacitor it can also follow a 50 ms pulse, and its larger gain resistors would give about three times finer current resolution (512 kΩ against 160 kΩ). The HSTIA was nevertheless preferred for practical reasons. Its feedback capacitor C_f is set in firmware [39, p. 52], whereas the LPTIA uses an external capacitor fixed at assembly, so the compensation can be changed without modifying the board. Its bandwidth also covers impedance measurement up to 200 kHz, against 300 Hz for the low-bandwidth loop [39, p. 9], which leaves impedance spectroscopy open as future work. On its most sensitive range the HSTIA resolves 0.23 nA per ADC code (Table D.3), and averaging 18 samples improves this further (Section 5.4). This meets Objective 2, but the LPTIA would be the better choice if the device were extended to sub-µg/L concentrations without preconcentration.

3.2.2 Stability with a Cell Connected

The AD5941 applies a voltage to the connected cell through feedback. The potentiostat amplifier drives the CE until the RE equals V_{BIAS}, the HSTIA holds the WE at V_{ZERO}, and the current is whatever the cell draws at that potential, up to 750 µA [39, p. 9]. The cell's electrical properties affect how both amplifiers behave. In the control loop, R_u and C_{dl} add a pole at $1/(2\pi R_u C_{dl})$. The gain-bandwidth product of this amplifier is fixed in the AD5941, so this loop cannot be compensated and instead limits the cells that can be driven [19].

The TIA loop can be shaped through C_f. Seen from the inverting input of the HSTIA, the cell is C_{dl} in series with $R_\Sigma = R_u + R_L$ to the solution, which the control loop holds

at a fixed potential. Without C_f the noise gain is

$$NG(s) = 1 + \frac{R_{\mathrm{TIA}}}{R_\Sigma + 1/sC_{dl}} = \frac{1 + s\,(R_{\mathrm{TIA}} + R_\Sigma)\,C_{dl}}{1 + sR_\Sigma C_{dl}}, \tag{3.1}$$

with a zero at $f_z = 1/2\pi(R_{\mathrm{TIA}} + R_\Sigma)C_{dl}$ and a pole at $f_p = 1/2\pi R_\Sigma C_{dl}$, above which it levels off at $1 + R_{\mathrm{TIA}}/R_\Sigma$. Unlike the photodiode amplifier often used as a model for this circuit [28], whose noise gain keeps rising, R_Σ isolates C_{dl} at high frequency. The loop crosses over where the open-loop gain of the amplifier, GBW/f, has fallen to the noise gain. With no cell connected the noise gain is 1 and the crossover is at the full GBW; with the electrode connected the crossover moves down to the plateau, $f_c \approx \mathrm{GBW} \cdot R_\Sigma/(R_{\mathrm{TIA}} + R_\Sigma)$. For a single-pole amplifier with $f_c \gg f_z$, the phase margin is approximately [40]

$$\phi_{\mathrm{PM}} \approx \arctan\left(\frac{f_c}{f_p}\right), \tag{3.2}$$

so a margin of at least 45° requires $f_c \geq f_p$, or

$$R_\Sigma \geq \sqrt{\frac{R_{\mathrm{TIA}}}{2\pi\,\mathrm{GBW}\,C_{dl}}}. \tag{3.3}$$

The margin is smallest on the largest gain resistor with the smallest series resistance. The datasheet does not give the GBW of the HSTIA, so a conservative 1 MHz is assumed; a larger value would only increase the margins. Table 3.1 gives the margin on the 160 kΩ range, computed from the full loop gain. The full derivation is given in Appendix D.13.

Table 3.1: Phase margin of the TIA loop on the 160 kΩ range ($R_\Sigma = R_L = 110\,\Omega$, $R_u = 0$, GBW = 1 MHz).

C_{dl}	10 nF	100 nF	1 µF	2.5 µF	5 µF
$C_f = 0$	5°	13°	38°	55°	69°
$C_f = 31$ pF (used)	22°	18°	40°	57°	70°
$C_f = 63$ pF	40°	24°	42°	58°	71°

With $C_f = 31$ pF on every range, the C220AT ($C_{dl} \approx 1.3$ µF to 5 µF, estimated from a typical 10 µF cm^{-2} to 40 µF cm^{-2} [19, Sec. 1.2] over its 0.126 cm^2 working electrode) has a margin of at least 44° with no solution resistance and 70° with $R_u = 100\,\Omega$, and the dummy cell WE A of Section 6.3 has 53°, consistent with its stable readings. Margins below 20° arise only near 100 nF on the sensitive ranges, where the TIA rings briefly after each step. With $\zeta \approx \phi_{\mathrm{PM}}/100$ and $t_s \approx 4.6/(\zeta \cdot 2\pi f_c)$ [40], the worst case settles to within 1 % in about 1 ms to 2 ms, long before the averaging window opens. Raising C_f to 63 pF would improve these margins with only a firmware change and is recommended as future work (Section 8.2).

Chapter 4

Hardware Design

This chapter details the hardware design focused on ensuring that the prototype functions correctly and to an adequate standard in its physical realisation by justifying the hardware selected. A primary concern was minimising the noise coupled into the measurement from other parts of the system, such as the Wi-Fi transmissions, the display, and reads and writes to the SD card. This was addressed through decoupling (Section 4.4.1), a ferrite bead on the analog supply (Section 4.4.2), and the separation of the analog and digital grounds (Section 4.4.3).

4.1 AFE Requirements

The research group at the University of Johannesburg [41] specified the measurement method as well as the parameters that the OpenStat should fulfil. The operating requirements are listed in Table 4.1. The same parameters are used on the PalmSens4 reference instrument in Chapter 6.

Table 4.1: Target DPV operating conditions and the requirements they impose.

Parameter	**Value**	**Requirement on the device**
Potential window	0.8 V to 1.3 V	Potential covers 0.8 V to 1.325 V, including the pulse
Step potential	5 mV	DAC LSB $\leq$ 1 mV, each level within $\pm$0.5 LSB
Pulse amplitude	25 mV	25$\pm$1 mV
Pulse time	50 ms	Timing error changes the sampled current by $\leq$ 2 %
Interval time	0.5 s	0.5 s $\pm$ 1 % per step, held over the full scan
Current range	$\pm$750 µA	No saturation at $\pm$750 µA, gain error $\leq$ 1 % per RTIA range after calibration

The AD5941 was specified at the start of the project and meets these requirements on a single chip, with 0.54 mV DAC steps and eight TIA gain ranges covering $\pm$750 µA [39]. Appendix D.15 compares it with the alternatives.

4.2 System Requirements

The table below details the requirements that the OpenStat must meet.

Table 4.2: System requirements.

ID	Requirement	Rationale
Processing and memory		
R1	Buffer a complete scan in memory	Samples are written to storage only after the scan finishes
R2	At least two SPI buses available for peripherals	One dedicated to the AD5941, one shared by the display and SD card
R3	Dedicated GPIO for the AD5941, display, SD card and user inputs	All peripherals connected without sharing control lines
Display and storage		
R4	Colour display with an SPI or I^2C interface	Compatible with the microcontroller
R5	Display fits in landscape orientation	Handheld form factor
R6	At least 100 horizontal pixels	One pixel per point of a 100-point scan
R7	Non-volatile storage	Field measurements accessed later
User interface		
R8	Standalone operation without a computer	Scan parameters set and scans started in the field
Power		
R9	Battery powered	Portable field use
R10	Rechargeable from USB	No battery replacement needed
R11	Regulated supply over discharge range	Cell voltage varies from 3.0 to 4.2 V
R12	At least 8 h of operation per charge	8-hour working day

To start implementing the above requirements, the microcontroller is specified next.

4.3 Microcontroller

The microcontroller must drive the AD5941 on a dedicated SPI bus while a second bus serves the display and SD card (R2), buffer a complete scan (R1), and serve the web interface over Wi-Fi without a companion app (Objective 4). The ESP32 family was chosen, as it is the MCU used by the designs surveyed in Table 2.2 that has native Wi-Fi that Objective 4 requires. Its newer ESP32-S3 member was used, on a development board based on the ESP32-S3-WROOM-2-N32R16V module [42]. This provides the two general-purpose SPI controllers needed for R2, and its 16 MB of PSRAM is far more than one scan requires, but leaves room to buffer several scans and the web assets.

4.4 PCB Component Placement and Layout

The key to low noise in the AD5941 is sufficient filtering and decoupling, together with correct component placement. The rules followed when selecting the decoupling capacitors were taken from the Analog Devices MT-101 tutorial [43]. Finally, a ferrite bead is placed in series with the supply pin. This keeps external high-frequency noise away from the IC and prevents internally generated noise from affecting the rest of the system [44].

4.4.1 Capacitor Choices

The capacitors around the AD5941 can be separated into two groups, decoupling the external supply pins and connecting internal nodes with the IC. Table 4.3 summarises the capacitor choices for their respective roles. The supply-pin values follow the MT-101 two-tier convention [43] and are checked against a target supply impedance, while the internal-node values follow the datasheet [39]. The full derivations for capacitor choices are given in Appendix D.9.

Table 4.3: Capacitors around the AD5941, grouped by the reasoning behind each value.

Pin	Value	Role and justification
DVDD (6), ×2 IOVDD (26) AVDD (30) AVDD (41)	100 nF	*High-frequency supply decoupling.* No datasheet value given, so one ceramic capacitor per supply pin, as per MT-101 [43]. Stays below the 1 Ω target impedance to beyond 100 MHz (Eq. (D.7)) [45].
AVDD (30)	10 µF	*Bulk decoupling.* Second tier of MT-101 [43]. Meets the 1 Ω target above 15.9 kHz (Eq. (D.5)) and forms an LC filter with the ferrite bead (Section 4.4.2).
`VREF_2V5` (4) `VBIAS_CAP` (31) `DVDD_REG_1V8` (14) `AVDD_REG` (5)	470 nF	*Reference and internal LDO decoupling.* Datasheet value [39, pp. 11, 20–21], set by the stability of on-chip buffers and regulators [46]. No value is given for `AVDD_REG`, so it is matched to `DVDD_REG_1V8`.
`VREF_1V82` (43)	4.7 µF	*ADC reference reservoir.* The datasheet recommends 4.7 µF [39, p. 19] but tests at 0.47 µF [39, p. 11]. The larger value was chosen as the more conservative.
`VBIAS0` (8) `VZERO0` (9)	100 nF	*Low-power DAC output noise filter.* Datasheet value [39, p. 20].
`RESET` (22)	100 nF	*Reset debounce.* With the 10 kΩ pull-up, $\tau = 1\,\text{ms} \gg 1\,\mu\text{s}$ minimum reset pulse [39, p. 14].

Together, the two tiers of supply capacitors hold AVDD below the 1 Ω target from about 16 kHz to beyond 100 MHz, which keeps supply noise below one LSB at the ADC reference. Noise arriving from the rest of the board is blocked separately by the ferrite bead.

4.4.2 Ferrite Bead

AVDD shares the 3.3 V rail with the ESP32-S3, display and SD card. These components place noise on the rail from tens of kilohertz to hundreds of megahertz [44]. This noise comes from the SPI clocks, regulator ripple and current bursts. A ferrite bead in series with AVDD forms a low-pass filter with the AVDD decoupling capacitors of Section 4.4.1, and over its resistive band it dissipates this noise as heat instead of reflecting it [44]. A bead was preferred over a separate regulator for AVDD because it adds a single small, inexpensive part that is most effective at high frequencies. The Würth Elektronik 74279204 was chosen for its impedance of 600 Ω at 100 MHz, with a rated current of 200 mA and a maximum DC resistance R_{DC} of 0.35 Ω [47].

The bead operates far from the point where DC current degrades its filtering. Impedance curves from the datasheet are measured at zero bias, and under load the core saturates and the inductance falls. Therefore, the bead should operate at no more than about 20 % of its rated current [44]. The worst-case AVDD current of about 10 mA is 5 % of the rating, so the datasheet impedance applies. The resulting 3.5 mV drop uses 1.2 % of the 0.3 V allowed between AVDD and DVDD [39, p. 7].

The primary risk with ferrite beads is resonance. Below its crossover frequency the bead is almost purely inductive, and it resonates with the decoupling capacitors. An underdamped resonance amplifies noise instead of attenuating it, which is called peaking [44]. For the series RLC filter,

$$f_0 = \frac{1}{2\pi\sqrt{L_{\mathrm{bead}}C}} \approx 50\,\mathrm{kHz}, \qquad Q = \frac{1}{R_{\mathrm{DC}}}\sqrt{\frac{L_{\mathrm{bead}}}{C}} \approx 0.90, \tag{4.1}$$

with $C = 10\,\mu\mathrm{F}$, the AVDD bulk capacitor of Table 4.3, and an assumed $L_{\mathrm{bead}} \approx 1\,\mu\mathrm{H}$, consistent with the 600 Ω rating at 100 MHz. A second-order filter peaks for $Q > 1/\sqrt{2}$ [40], so the response rises by about 0.7 dB near 30 kHz, which is negligible.

It is the 10 µF bulk capacitor that keeps the resonance well damped. The large capacitor lowers $\sqrt{L/C}$ until the bead's own series resistance is sufficient to damp the filter, so the bulk capacitor also serves to damp the bead.

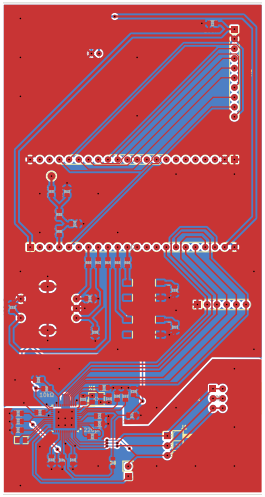
At high frequencies the filter performs as intended. At 100 MHz the bead's 600 Ω and the 0.63 Ω of the local 100 nF capacitors (Eq. (D.7)) form a divider that attenuates noise by about $20\log_{10}(0.63/600) \approx -60$ dB.

4.4.3 Analog and Digital Ground

The PCB carries two types of signals, analog and digital, whose requirements conflict. The AD5941 AFE measures small Faradaic currents, so any disturbance appears as a measurement error. In contrast, the ESP32-S3, the display and the user inputs are digital. This means that they draw fast pulses of current at set logic levels, and that current must return to its source through ground. A ground plane is not ideal for this as it has a small resistance and inductance, so a rapidly changing current induces a voltage according to $v = L\,\mathrm{d}i/\mathrm{d}t$. If the analog and digital circuits share the same ground return, this voltage is added to the analog reference and the digital switching noise is superimposed on the measured signal [48].

To combat this, the PCB was partitioned into an analog region and a digital region, which are the lower and upper parts of the board respectively (Figure 4.1), with all digital components and traces confined to the digital region. Partitioning is effective because, at high frequencies such as those in digital edges, the return current follows the path of least inductance and flows in the ground copper directly beneath its own signal trace [49, 50]. In accordance with Analog Devices guidance, the analog and digital grounds should be joined through the shortest possible connection [48, 50]. This is done by stitching them together directly beneath the AD5941, commonly called a star ground connection [48]. The stitched bridge also serves as the crossing point for the SPI bus and the reset and interrupt lines, so each signal retains a continuous return path directly underneath it. If the two grounds had been fully separated and joined only at a distant point, such as the power input, these return currents would be forced around a large loop that acts as an antenna [51].

Figure 4.1: PCB copper layout, with the digital region at the top and the analog region at the bottom.



At the sub-nanoampere currents estimated in Section 2.1.4, leakage and interference on the working-electrode path can set the noise floor. The electrode lead used in testing was unshielded; however, it was kept short to minimise noise pickup. A spare analog ground pin beside the electrode interface allows a shielded cable to be terminated at the interface.

4.4.4 Trace Widths

Trace widths were checked with the IPC-2221 equations [52] as implemented in the DigiKey calculator [53], for a 10 °C rise in 1 oz/ft^2 external copper. The 0.3 mm traces used for the AD5941 and display nets can carry 1.0 A and the 0.6 mm supply traces 1.65 A, well above their design currents of 10 mA, 33 mA and 1 A. See Appendix D.14 for more information.

4.5 Display and SD Card

The DFRobot Fermion 1.8-inch TFT (DFR0928) was selected as the display because it meets requirements R4–R7 from Table 4.2. The Fermion was the only component in stock that met all the criteria. Additional information regarding the display is given in Appendix Table D.1.

The display and the SD card share a single SPI bus, because the ESP32-S3 has only two general-purpose SPI controllers [42]. This allows a full SPI bus to be dedicated to the AD5941, so that scans are minimally affected. The cost is that the display and the SD card cannot be used at the same time. This is avoided by buffering the samples in PSRAM and writing the file only after the scan has finished.

4.6 Power Management

The power subsystem was chosen to meet the requirements of 8-hour working day battery life, over-voltage and under-voltage protection, and affordable components. The chosen components are the TP4056 charger with a DW01A protection IC, the Pololu S9V11E2F5 step-up/step-down regulator producing 5 V, and a 5000 mA h, 3.7 V single-cell LiPo battery. Their main specifications are summarised in Table D.2.

4.6.1 Battery

To verify that the selected 5000 mA h battery fulfils the full-day operational requirement, the system's current was measured during idle, transmission over Wi-Fi and DPV measurements. The supply current was measured by powering the PCB at 5 V from a Siglent SPD3303C bench supply and reading the output current from the supply. The SPD3303C has a current readback resolution of 10 mA and an accuracy of ±0.5 % [54], that is, about ±21 mA at these currents. The board draws 100 mA ± 21 mA at idle and 160 mA ± 21 mA while transmitting over Wi-Fi. The idle current did not change when taking a DPV measurement. Taking 160 mA as the worst case, the power drawn from the battery through the regulator ($\eta_{\text{reg}} \approx 0.9$) is

$$P_{\text{bat}} = \frac{5\,\text{V} \times 160\,\text{mA}}{\eta_{\text{reg}}} = \frac{0.80\,\text{W}}{0.9} \approx 0.89\,\text{W}, \tag{4.2}$$

and the run time is

$$t_{\text{run}} = \frac{E_{\text{bat}}\,\eta_{\text{reg}}}{V_{\text{out}}\,I_{\text{load}}} = \frac{5\,\text{A h} \times 3.7\,\text{V} \times 0.9}{5\,\text{V} \times 160\,\text{mA}} \approx 21\,\text{h}. \tag{4.3}$$

Allowing for the measurement uncertainty, the run time with Wi-Fi transmitting lies between 18 h and 24 h, and at idle between 28 h and 42 h. Even the lower bound comfortably meets R12, as well as R9–R11.

4.6.2 Charging

The TP4056 is a linear constant-current, constant-voltage charger with a 4.2 V float voltage, which charges at up to 1 A [55]. A full charge of the 5000 mA h cell was measured to take about 6.5 h, longer than the 5 h that capacity divided by current suggests, because the charger reduces the current in its final constant-voltage stage.

4.6.3 Regulation

The Pololu S9V11E2F5 covers the full battery voltage range with sufficient output current, but it requires the addition of an external low-voltage cutoff. An MT3608 boost converter was also considered, but was rejected after three of the modules tested proved faulty. The Pololu's 700 mA start-up current limit is sufficient for the ESP32-S3 to boot, and its soft-start ramps the output up gradually [56]. The regulator has a 100 kΩ pull-up resistor from its enable pin to V_{IN}; the enable pin turns the regulator off below 0.7 V and on above 0.8 V [56]. Connecting a 33 kΩ resistor from the enable pin to ground forms a divider that switches the regulator off at about 2.8 V, as given in Equation (4.4).

The divider draws about 28 µA continuously, which is sufficiently low for this application.

$$V_{\mathrm{off}} = 0.7\,\mathrm{V} \times \frac{100\,\mathrm{k\Omega} + 33\,\mathrm{k\Omega}}{33\,\mathrm{k\Omega}} \approx 2.8\,\mathrm{V} \tag{4.4}$$

The 5 V output supplies the ESP32-S3 development board, whose on-board regulator provides the 3.3 V rail for the whole PCB, including AVDD of the AD5941. A design oversight was the choice to boost the voltage to 5 V when no component requires a 5 V supply. A 3.3 V step-up/step-down regulator would have been more efficient. The large 5000 mA h battery makes this loss affordable, although it is not ideal.

4.6.4 Input Decoupling

No single capacitor has a low impedance across the whole frequency range, so a large capacitor for low frequencies is paired with a small one for high frequencies [43]. Decoupling is required because high-frequency noise couples through to the regulator output and must be kept out of the chip [43]. A 100 µF bulk capacitor is used at the output of the boost regulator, which is the input to the PCB, to act as a local charge reservoir. An additional 100 nF capacitor shunts high-frequency noise away from the PCB [43].

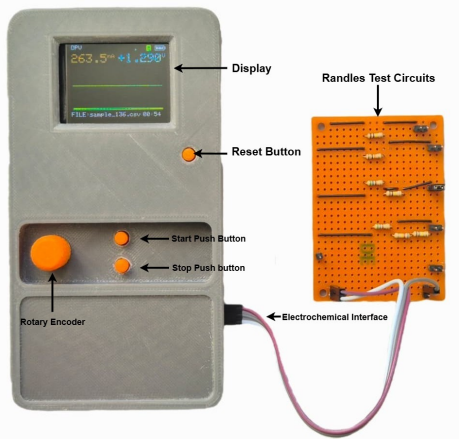
4.7 User Interface

The interface consists of two push buttons and a rotary encoder with an integrated push button. All are wired in an active-low configuration with 10 kΩ hardware pull-up resistors. A capacitor on each line forms an RC filter with its pull-up, so on release the pin only reads high once the capacitor has charged past the ESP32-S3 input-high threshold of $0.75 \times V_{DD}$ [57]. This takes $t = RC \ln 4 \approx 1.39\,RC$.

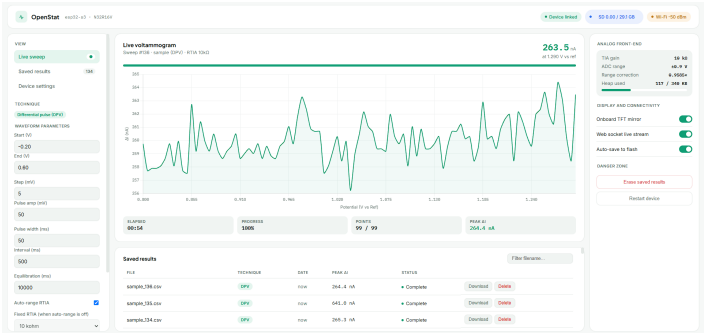
The push buttons use 100 nF (1.4 ms) to absorb contact bounce [58], and the encoder lines use 10 nF (0.14 ms) so that fast rotation is not smeared; the remaining encoder bounce, which can last up to 2 ms [59], is rejected by the quadrature decoder in firmware.

The OpenStat has two user interfaces, shown in Figure 4.2. On the device, the rotary encoder selects and edits the scan parameters of Table 5.1, the push buttons start and stop a scan, and the display shows the voltammogram as it is measured. The web interface offers the same parameters, a live voltammogram and a list of stored scans that can be downloaded. If the device cannot join a known Wi-Fi network, it starts its own access point, so that a phone or laptop can connect to it in the field (Section 5.2).

Figure 4.2: OpenStat user interfaces. (a) Physical interface, shown connected to the Randles test circuits. (b) Web interface displaying a live voltammogram.



(a)



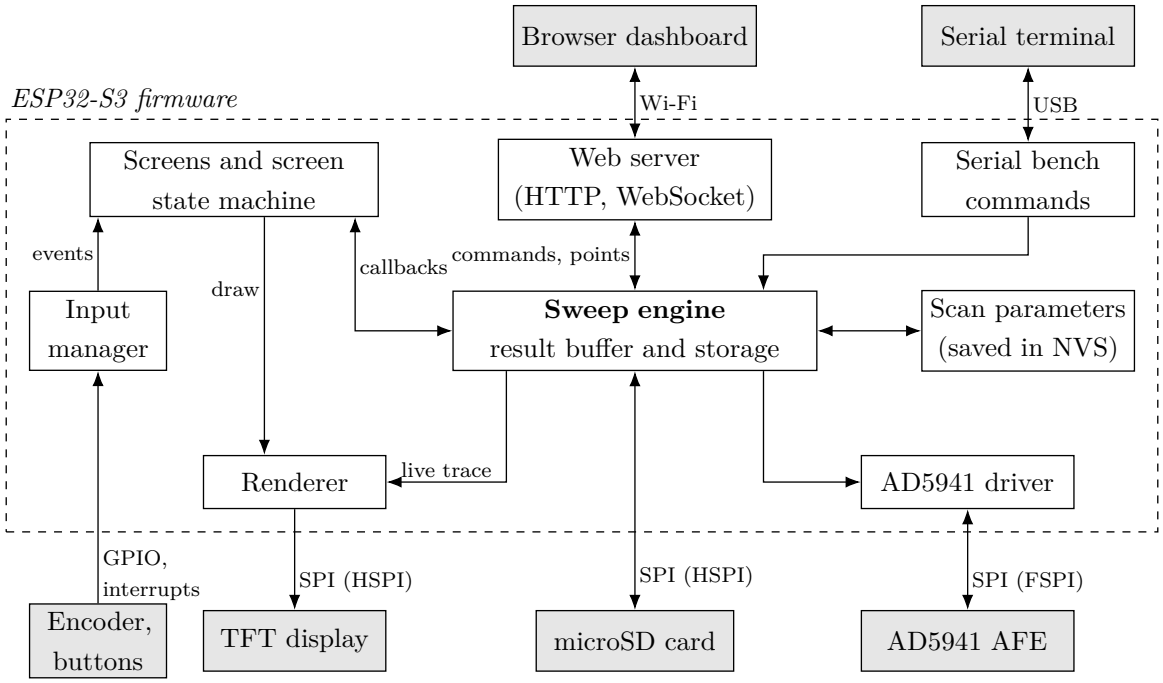
(b)

Chapter 5

Firmware Design

The hardware described in Chapter 4 has the ability to apply a potential to the cell and measure the current that flows, but it is the firmware that turns these two features into a useful measurement. This chapter describes the structure of the firmware and the logic of how a measurement is performed.

Figure 5.1: Firmware structure. Arrows show the direction of calls or data.



The modules in Figure 5.1 are described in the sections that follow.

5.1 Firmware Overview

The firmware is written in C++ on the Arduino framework for the ESP32-S3 and is built with PlatformIO. Third-party libraries like Adafruit GFX and ST7735 are used for the display, ESPAsyncWebServer and AsyncTCP for the web server and ArduinoJson for JSON encoding.

Three design decisions shaped the structure of the firmware:

- **Project-specific AD5941 driver.** Analog Devices supplies a general-purpose firmware library for the AFE, which covers every measurement mode of the device. The driver for this project was adapted from this library and was written around the needs of this project. This gives the firmware direct control of the gain resistor R_{TIA} of the transimpedance amplifier, which the automatic range selection in Section 5.5.2 switches repeatedly before a scan.
- **One sweep engine, two front ends.** The on-device screens and the web dashboard both start, stop and observe a measurement through the same sweep engine, so a scan behaves identically regardless of the interface that started the sweep.
- **A dedicated SPI bus for the AFE.** The display and the SD card share the second SPI bus (Section 4.5), so display and storage traffic never occupies the bus used for the measurement.

5.2 Top-Level Program Flow

All project code runs from a single loop, shown in Figure 5.2. During start-up the device checks that the AD5941 and SD card are working and connects to Wi-Fi, starting its own access point if no known network is found, before opening the home screen. The state of the device is stored in three places: the active screen, a flag that records whether a scan is running, and the phase of the DPV step inside the AFE driver. The loop services each of these in turn on every pass and none of them blocks the loop, except when the device is sampling which is discussed in Section 5.4.

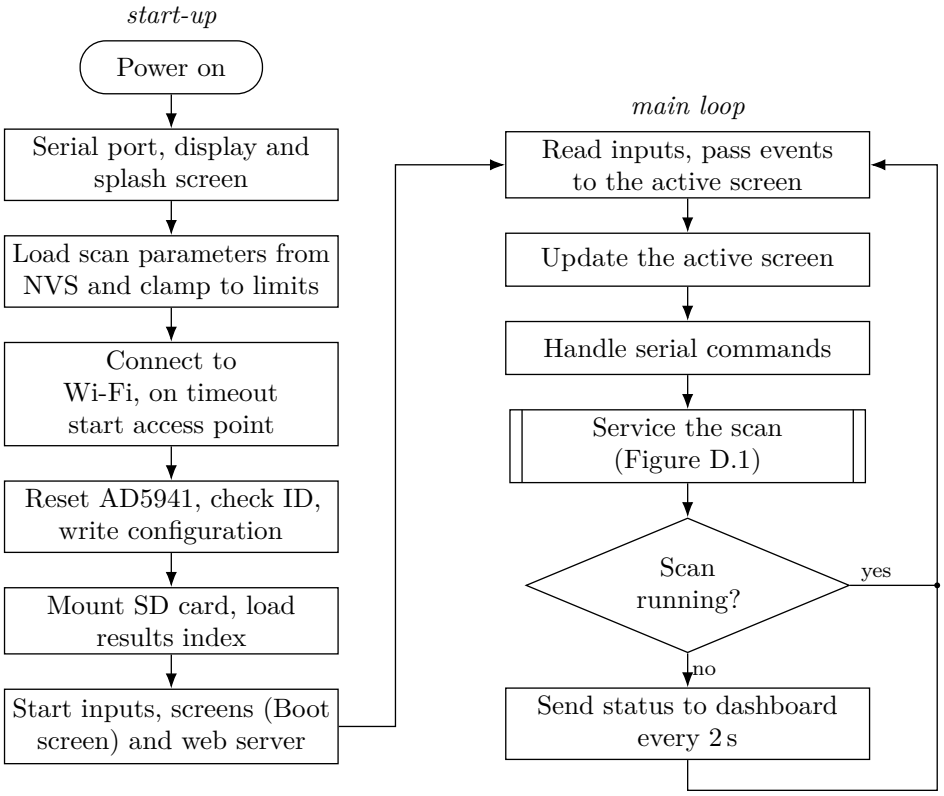
5.3 DPV Measurement Sequence

A scan is defined by the parameters in Table 5.1. They are stored in non-volatile storage (NVS), a partition of the ESP32-S3's flash memory, edited on the device and clamped to these limits whenever they are loaded or edited, to protect the hardware. The start potential may lie above or below the end potential; the firmware takes the direction of the staircase from their order, so both the ascending scans of Section 6.3 and the descending scans of Sections 6.4 and 6.5 are supported. The scan rate is not set directly; it is $\Delta E_{\mathrm{s}}/T$, which is $10\,\mathrm{mV\,s^{-1}}$ for the default values.

5.3.1 Generating the Potential

The cell potential is set by the two outputs of the low-power DAC of the AD5941. The 6-bit output V_{ZERO} biases the high-speed transimpedance amplifier (TIA), which holds

Figure 5.2: Top-level program flow, start-up sequence (left) and the cooperative main loop (right).



the working electrode at that voltage, and is fixed at 1.7125 V. The 12-bit output V_{BIAS} drives the reference electrode through the potentiostat amplifier and carries the staircase and the pulses. The applied potential is therefore

$$E = V_{\mathrm{ZERO}} - V_{\mathrm{BIAS}}, \qquad V_{\mathrm{BIAS}} = 0.2\,\mathrm{V} + D_{12} \times \frac{2.2\,\mathrm{V}}{4095}, \tag{5.1}$$

where D_{12} is the 12-bit code; one code is 537.2 µV. With V_{ZERO} fixed at this value, the permitted range of E in Table 5.1 keeps V_{BIAS} between 0.41 V and 2.06 V, inside the 0.2 V to 2.4 V output range of the DAC.

Table 5.1: DPV scan parameters, their default values and permitted ranges.

Parameter	Symbol	Default	Permitted range
Start potential	E_{start}	0.80 V	−0.35 V to 1.30 V
End potential	E_{end}	1.30 V	−0.35 V to 1.30 V
Step	ΔE_{s}	5 mV	1 mV to 50 mV
Pulse amplitude	ΔE_{p}	25 mV	−100 mV to 100 mV
Pulse width	t_{p}	50 ms	25 ms to 500 ms
Pulse period	T	500 ms	50 ms to 5000 ms
Equilibration time	t_{eq}	10 s	configurable
TIA range	R_{TIA}	automatic	automatic, or one of eight fixed values

5.3.2 Scan Flow

Before any timing starts, the TIA range is selected (Section 5.5.2) and the requested timing is checked against the minimum that the sampling scheme can support (Section 5.4); a scan that fails this check is refused. The cell is then held at E_{start} for the equilibration time t_{eq} before the first step, so that the charging transient of the large initial step from 0 V decays; the same 10 s was used on the PalmSens4 in Chapter 6. The scan consists of

$$N = \left\lfloor \frac{|E_{\text{end}} - E_{\text{start}}|}{\Delta E_{\text{s}}} \right\rfloor \tag{5.2}$$

points, and the end potential itself is not applied as a base level. Each point is recorded at its base potential, calculated from the DAC code actually written rather than the requested value. Each point is produced by the step routine in Figure D.2, which the main loop calls on every pass. The routine returns immediately unless a sampling window is due, so the user interface stays responsive during the 450 ms base dwell.

Points are held in memory during the scan and written to the SD card as a single file only once the scan has stopped, so that card activity cannot disturb the measurement. Every way of leaving a scan, whether completion, a stop request or an overload, passes through the same stop routine, which stops the ADC, returns the cell to $E = 0$ V and saves the points captured so far. Figure D.1 (Appendix D.4) shows the flow of one scan.

Each scan is saved as a comma-separated file with one row per point (E, I_{before}, I_{end}, ΔI, range code and time stamp), and an index file lists the saved scans. The dashboard exchanges JSON messages with the firmware over a WebSocket, so points are plotted as they are measured, and saved files are downloaded over HTTP.

5.4 Pulse Sampling and Timing

DPV rejects the capacitive charging current only if I_{before} and I_{end} are each sampled after the transient from the preceding edge has decayed, as shown in Section 2.1.3. When the current is sampled, and for how long, therefore sets both the noise of the measurement and how much charging current leaks into ΔI.

5.4.1 Firmware Timing

The AD5941 contains a programmable sequencer that can apply DAC updates and start ADC conversions from its own clock, without the microcontroller [39]. It was not used. Instead, the staircase and the pulses are individual DAC writes over SPI, scheduled by the ESP32-S3 against its millisecond clock, and data-ready is detected by polling the interrupt

flag register of the AD5941. This was a deliberate trade-off; the sequencer would time each pulse more precisely, but timing the measurement in firmware gives the ESP32-S3 full control over when every operation in the system takes place, so the measurement can be scheduled around the display, storage and network activity instead of running alongside it. The cost is measured in Section 6.1.2 where the pulse is on average 3 % shorter than set, which changes the sampled current by about 1.5 %, within the 2 % allowed by Table 4.1.

The ESP32-S3 controls the flow to give the measurement priority over everything else while a scan is running. Whenever the current is being sampled, the main loop does nothing but sample. This keeps the digital activity of the device out of the sampling windows, which reduces the noise coupled into the measurement and improves its accuracy. The cost is that the timing between windows depends on the main loop, and the sampling scheme below was designed so that this dependence does not reach the measured current.

5.4.2 Averaging Window

Before the ADC of the AD5941 transmits the measurements to the ESP32-S3, its output passes through two digital filters inside the device, a sinc3 filter followed by a sinc2 filter [39, p. 57]. A sinc filter is a low-pass filter that averages a block of consecutive samples and outputs one result per block, so it both suppresses noise and lowers the data rate. The number of samples per output is the oversampling ratio. In the configuration used, the sinc3 filter reduces the raw 800 kHz ADC rate by a factor of 5 to 160 kHz, and the sinc2 filter reduces it further by a factor of 178 [39, p. 59], resulting in an effective sampling frequency of 900 Hz. The digital bandwidths of the sinc3 and sinc2 filters are about 42 kHz and 290 Hz. The device also offers a 50 Hz/60 Hz notch filter after the sinc2 filter, but with it enabled the settling time after a step is 37 ms, against 2.245 ms without it [39, p. 17]. A 37 ms settling time would consume most of a 50 ms pulse, so the notch filter is bypassed and mains rejection is provided in firmware instead. Each current value is the mean of $N_{\mathrm{w}} = 18$ consecutive sinc2 results. The window length is

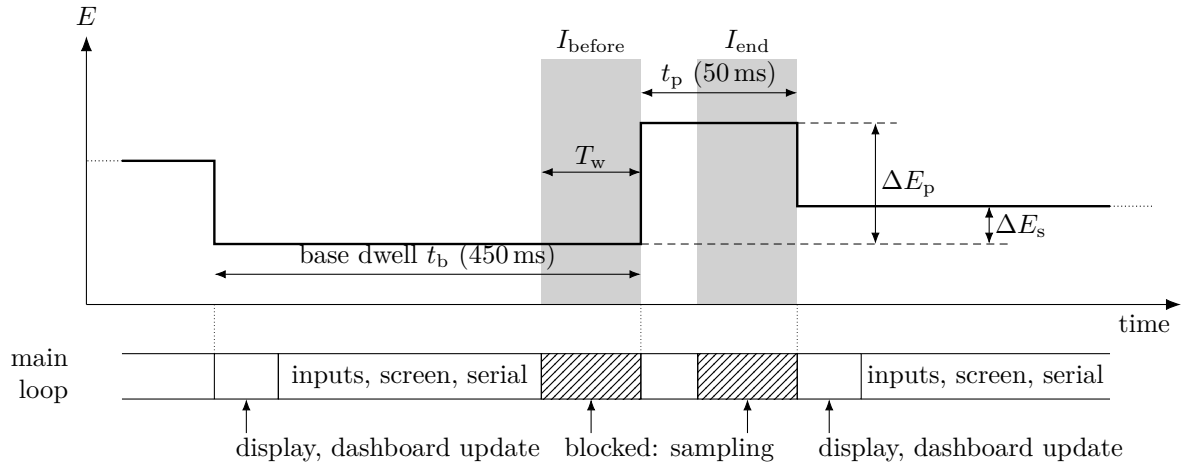
$$T_{\mathrm{w}} = N_{\mathrm{w}} T_{\mathrm{s}} = 18 \times 1.1125\,\mathrm{ms} = 20.025\,\mathrm{ms}, \tag{5.3}$$

which is one period of the 50 Hz mains to within 0.13 %. The mean has zeros at multiples of $1/T_{\mathrm{w}} = 49.94\,\mathrm{Hz}$, so mains interference and its harmonics are rejected. The window is defined by a sample count and not by a duration, so every window in a scan has exactly the same length, and the ADC runs continuously for the whole scan so that the filters are never restarted between windows.

5.4.3 Placing Windows

Figure 5.3 shows where the two windows fall within a step. Each window starts 21 ms before the edge that follows it, so that it ends at the edge. I_{before} is the mean current over the last 20 ms of the base dwell and I_{end} the mean over the last 20 ms of the pulse. These are the points of each phase at which the charging current has had the longest time to decay.

Figure 5.3: One DPV step with the default timing. Shaded bands show the two averaging windows.



Three properties of this scheme limit the effect of firmware timing:

- **During sampling the loop is blocked.** While a window is being sampled, the main loop waits on the data-ready flag of the ADC and does nothing else. This means that the 18 samples are consecutive, and no display, storage or serial activity takes place while the current is being sampled.
- **Lateness only adds settling time.** If a window starts late, the sampling starts later after the edge, so loop delay lengthens the pulse slightly but cannot move a window into the charging transient.
- **Slow work is placed after the step.** The display and the dashboard are updated when a point is returned, which is at the start of the next base dwell.

A scan is refused if the pulse width or the base dwell is shorter than the window plus a 5 ms margin, that is, 25 ms. At this minimum a window still starts 4 ms after the preceding edge, which is longer than the 2.245 ms settling time of the filter.

5.4.4 Limitations

Each window starts 21 ms before its nominal edge but lasts only 20.025 ms, and the next level is written as soon as the window ends, so every phase ends about 1 ms early; the millisecond timer adds up to a further 1 ms of truncation, and there is no correction against an absolute schedule. This is consistent with the measured pulse width and period, which are 3.0 % and 0.7 % shorter than set (Table 6.2). Starting each window 20 ms before the edge would reduce this error. The buttons are not read during a window, so a stop request can be delayed by up to 20 ms. Commands from the dashboard are handled in the callback context of the web server library and not in the main loop, so their timing relative to a window is not controlled.

5.5 Current Measurement and Ranging

The ADC measures the differential output of the high-speed TIA through its gain stage, set to a gain of 1.5. This is the only factory-calibrated gain [39, p. 56]. The mean code $\bar{D}$ of a window is converted to the TIA output voltage with the transfer function given in the datasheet for this gain [39, p. 56], and then to a current using the nominal value of the selected gain resistor:

$$V_{\mathrm{TIA}} = \frac{1.835\,\mathrm{V}}{1.5} \times \frac{\bar{D} - 2^{15}}{2^{15}}, \qquad I_{\mathrm{raw}} = \frac{V_{\mathrm{TIA}}}{R_{\mathrm{TIA}}}. \tag{5.4}$$

A gain and an offset correction, held in a table with one entry per range k, are then applied:

$$I = G_k\, I_{\mathrm{raw}} + O_k. \tag{5.5}$$

Both samples of a step are taken on the same range, so the offset cancels in the differential current and only the gain affects the result:

$$\Delta I = I_{\mathrm{end}} - I_{\mathrm{before}} = G_k \left(I_{\mathrm{raw,end}} - I_{\mathrm{raw,before}}\right). \tag{5.6}$$

5.5.1 Ranges and Calibration

The eight gain resistors of the high-speed TIA give the ranges from 200 Ω to 160 kΩ in Table D.3. The full-scale current follows from the ±0.9 V input range of the ADC at this gain [39, p. 8], and the resolution is the current that corresponds to one ADC code before averaging.

The potentiostat amplifier that drives the counter electrode can source or sink only ±750 µA in its normal mode [39, p. 9], which is the mode used. The 1 kΩ resistor covers this range

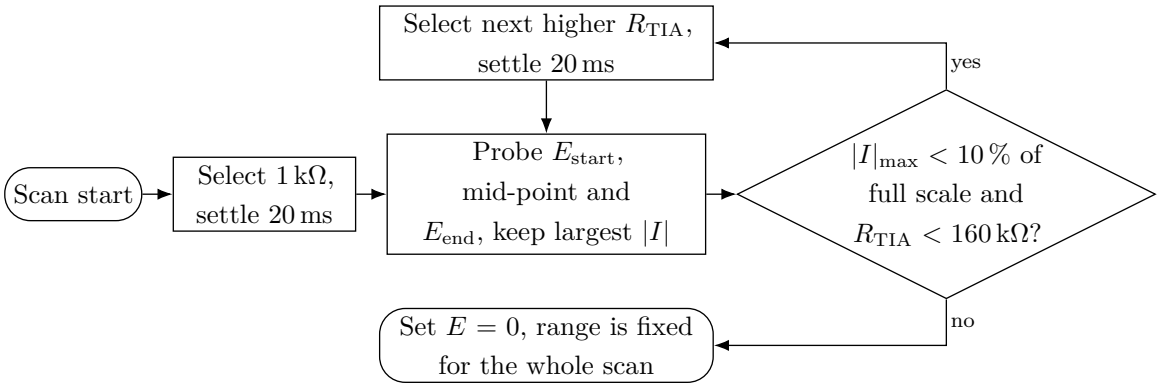
up to its overload threshold of 720 µA (Section 5.5.2). Currents between 720 and 750 µA therefore end a scan as overloaded, so the ±750 µA requirement of Table 4.1 is met only to 720 µA. The 200 Ω range would only become useful with the high-current mode of the amplifier enabled.

The AD5941's own RCAL calibration resistor (220 Ω, 0.1 %) is fitted but not used; instead, each range is calibrated against an external resistor connected in place of the cell (Section 6.2). The gain and offset constants are compiled into the firmware and are listed in Appendix D.5.

5.5.2 Automatic Range Selection

Before a scan the firmware selects the range with the procedure in Figure 5.4. It starts on the 1 kΩ range, which cannot be overloaded by any current the cell is expected to produce, and measures the current at the start, middle and end potentials of the scan with no pulse applied. If the largest of the three is below 10 % of full scale, the next more sensitive range is selected and the probe is repeated. Successive ranges differ by a factor of five or two, so a signal just below the threshold occupies at most 50 % of full scale on the range that is finally chosen. The remainder is headroom for the pulse response and the Faradaic peak, which the probe does not see because it applies no pulse. Automatic selection therefore uses the seven ranges from 1 kΩ to 160 kΩ; the 200 Ω range can only be chosen manually.

Figure 5.4: Automatic selection of the TIA range before a scan.



The range is then held for the whole scan. Changing the gain resistor during a scan would place the two samples of a step on different ranges, where the offsets no longer cancel in Equation (5.6), and would add a settling transient. If the window-mean TIA voltage exceeds 80 % of full scale (0.72 V) on either sample, the scan is instead stopped, the points captured before the overload are saved and the result is marked as overloaded. The user can also disable automatic selection and choose a fixed range.

Chapter 6

Testing and Calibration

This chapter evaluates whether the developed potentiostat matches the performance of a commercial one. Discrepancies with the reference potentiostat and their likely causes are discussed. To isolate error sources, the validation follows a sequence of five progressive tests. Each stage introduces an additional component of a complete electrochemical measurement and benchmarks it against the best available reference standard, allowing any observed deviation to be traced directly to the newly introduced variable. Each test is listed with the key performance parameter of Section 2.2.2 that it evaluates:

1. **Excitation waveform (technique support):** The potential steps, pulses and timing output by the AFE (Section 6.1).
2. **Resistors (current range and resolution):** The current measurement, compared with a digital multimeter (Section 6.2).
3. **Randles circuits (noise floor):** Dummy cells made from known components and compared against a commercial or laboratory grade instrument, like the PalmSens4 (Section 6.3).
4. **Salt solution (noise floor):** A commercial or laboratory grade electrode without a redox reaction, against the PalmSens4 (Section 6.4).
5. **Ferricyanide (technique support):** An active redox peak, against the PalmSens4 (Section 6.5).

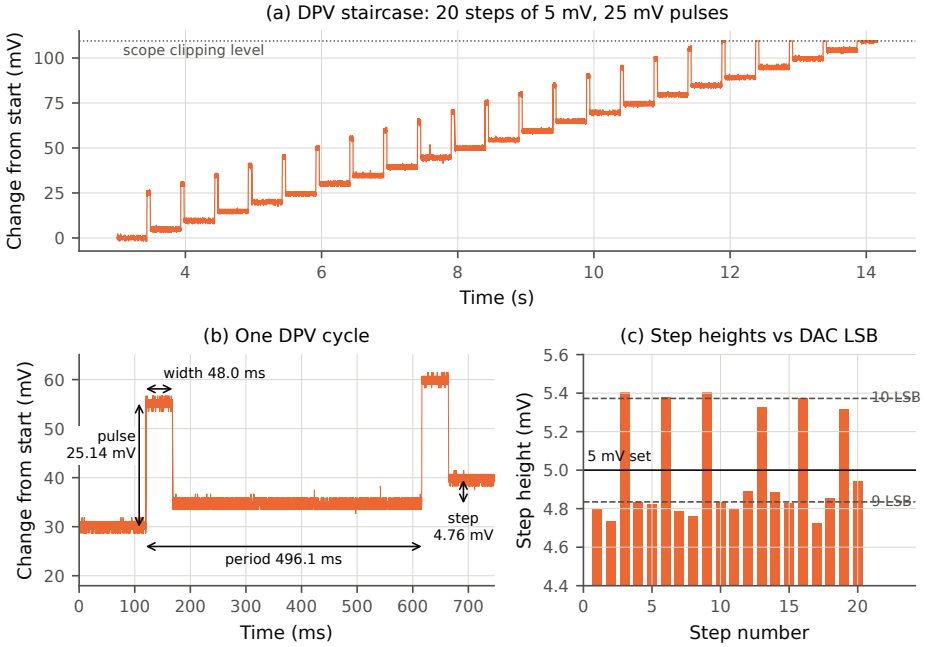
Table 6.1: Measurement conditions.

Electrode	Metrohm DropSens C220AT SPE: 4 mm gold working electrode, gold counter, silver pseudo-reference [60], reused after rinsing with ethanol
Solutions	Blank: 0.1 M KCl. Probe: 5 mM $K_3[Fe(CN)_6]$ in 0.1 M KCl. One drop covering all three electrodes
DPV (Randles)	0.8 V to 1.3 V, step 5 mV pulse 25 mV, 50 ms, interval 0.5 s
DPV (salt, FC)	0.6 V to −0.2 V (KCl) or −0.35 V ($K_3[Fe(CN)_6]$), step 5 mV, pulse 50 mV, 50 ms, interval 0.2 s; all tests 10 s equilibration on both instruments

6.1 Excitation Waveform

Every later test assumes that the DPV waveform is correctly generated. Hence, the first check is to confirm the output of the device before any current is measured. The excitation was recorded with a Siglent SDS1104X-E oscilloscope [61] while the device ran a DPV scan with a 5 mV step, a 25 mV pulse, a 50 ms pulse width and a 0.5 s interval. The results are shown in Figure 6.1 and Table 6.2.

Figure 6.1: Excitation waveform measured with the oscilloscope. (a) The complete staircase. (b) One cycle with the measured quantities. (c) Height of every step against 9 and 10 LSB of the AD5941 12-bit DAC.



6.1.1 Amplitude

Every step is either 4.76 mV or 5.37 mV and never deviates as illustrated in Figure 6.1c. These are 9 and 10 codes of the 12-bit DAC that sets the cell potential [39]. A 5 mV step is 9.31 LSB, so the firmware rounds each level to the nearest code, and the steps follow the pattern 9, 9, 10 with a mean of 4.98 mV. Because each level is computed from its absolute target rather than by adding steps, the rounding does not accumulate. The pulse is rounded to 47 LSB (25.25 mV) and measures 25.17 mV. The difference between the two step sizes, 0.545 mV, matches the LSB to within the ±3 % gain accuracy of the oscilloscope [61], which limits how precisely the amplitudes can be verified. The applied potential is therefore correct to within one LSB. A 0.54 mV resolution is more than enough for DPV, where a peak is normally about 90 mV wide [19].

6.1.2 Timing

The edges settle in less than 0.1 ms, which is negligible against the 50 ms pulse. The pulse width alternates between 48.0 ms and 49.1 ms, and the period between two values about 1 ms apart, consistent with the 1 ms resolution of the firmware timer. Both are shorter than set, the pulse by 1.5 ms on average (3 %) and the period by 3.5 ms (0.7 %).

Timing does not affect the peak potential, which identifies the compound, so these errors change only the peak height. For a diffusion-limited current, which decays as $t^{-1/2}$ [19], sampling 3 % early raises the measured current by about 1.5 %, and the 1.1 ms alternation varies it by about 1 % between points. The device therefore applies the programmed waveform, with amplitudes correct to one LSB and timing within 3 %. With the excitation verified, the next check is the current it measures.

Table 6.2: Measured excitation waveform against the programmed values.

Parameter	Set	Measured	Error
Step height (20 steps)	5 mV	(4.98 ± 0.26) mV	−0.3 %
Pulse height (17 pulses)	25 mV	(25.17 ± 0.08) mV	+0.7 %
Pulse width	50 ms	(48.5 ± 0.6) ms	−3.0 %
Period	500 ms	(496.5 ± 0.6) ms	−0.7 %
Edge time (10–90 %)	–	<0.1 ms	–

6.2 Resistor Calibration

The firmware converts the high-speed TIA output to a current using the value of the selected gain resistor R_{TIA} [39], so any difference between the nominal and the actual gain of the signal chain appears directly as a current error. Each of the eight ranges was therefore calibrated separately.

A Plusivo DM101 digital multimeter [62] was used to measure a resistor R, which was connected in place of the cell. On its 2 V DC range the multimeter has a resolution of 1 mV and an accuracy of ±0.5 %, and on its 2 kΩ to 200 kΩ resistance ranges an accuracy of ±0.8 %. The value of R was chosen so that ±0.5 V drives a current near the top of its operational range. At ±0.5 V the multimeter was used to measure the voltage V across the resistor, giving the true current $I_{\mathrm{true}} = V/R$, and the firmware reading I_{meas} was recorded. Denoting the two operating points by the subscripts + and −, a linear correction is fitted

through them:

$$I_{\text{true}} = k\, I_{\text{meas}} + c, \qquad k = \frac{I_{\text{true},+} - I_{\text{true},-}}{I_{\text{meas},+} - I_{\text{meas},-}}, \qquad c = I_{\text{true},+} - k\, I_{\text{meas},+}, \tag{6.1}$$

where the gain k corrects the scale and the offset c the zero. The two points lie at opposite ends of the range, so the slope can be estimated accurately. The correction was verified at $+0.25\,\text{V}$, a point not used in the fit.

Before calibration, I_{meas} exceeded I_{true} by 9 % to 12 % on all eight ranges (Table 6.3). An error of the same size and sign on ranges with different resistors points to a cause shared by the conversion chain, such as the assumed ADC reference or PGA gain, rather than to the tolerance of the external resistors, although the internal gain resistors of the AD5941 may also share a common process error. After correction the maximum error is 0.44 %.

Table 6.3: Two-point calibration of the high-speed TIA ranges, recomputed from the recorded readings. Errors are measured at $+0.25\,\text{V}$.

Range	R **(DMM)**	**Gain** k	**Offset** c **(nA)**	**Error before (%)**	**Error after (%)**
200 Ω	994 Ω	0.9013	160.0	11.36	+0.44
1 kΩ	994 Ω	0.9141	278.7	9.70	+0.40
5 kΩ	4.59 kΩ	0.9089	−69.5	9.99	−0.16
10 kΩ	9.93 kΩ	0.8971	−81.6	11.78	−0.05
20 kΩ	19.9 kΩ	0.9018	−15.8	11.09	+0.06
40 kΩ	38.5 kΩ	0.9129	−9.2	9.72	+0.02
80 kΩ	81.5 kΩ	0.9158	−3.0	9.44	+0.13
160 kΩ	146.7 kΩ	0.9191	1.1	9.15	+0.38

With the gain of every range corrected and checked against an independent reference, the current scale of the device is known to within about ±1 %, limited by the multimeter used as the reference. This is crucial from an application perspective because the drug concentration is calculated directly from the peak current. The next test uses Randles circuits and adds a capacitive load to the system.

6.3 Randles Circuit Test

The next test adds the capacitance of the cell. A real electrode has a double-layer capacitance, which the DPV sampling of Section 2.1.3 is designed to reject. The Randles test checks this with passive networks of known DC resistance in place of the cell [18]. A total of 7 test cells were used, three cells of the PalmSens4 dummy cell (WE A to C) and four networks built for this project (Cells 1 to 4), each scanned on both instruments. Once the capacitors have charged, the DPV current depends only on the DC path, so

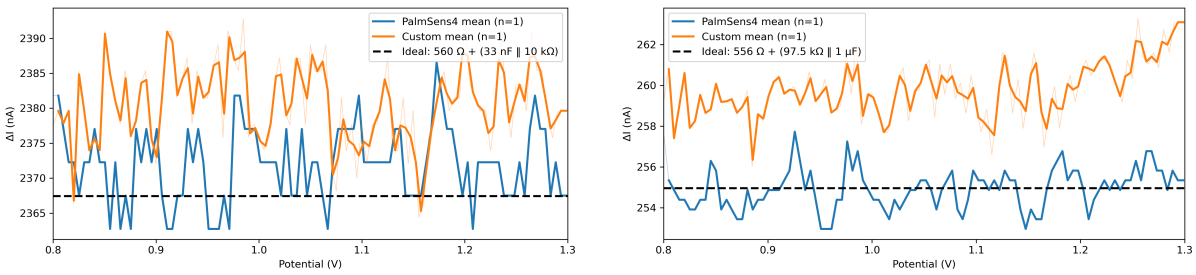
the expected value is $\Delta I = \Delta E_{\text{pulse}}/(R_s + R_{ct})$ with $\Delta E_{\text{pulse}} = 25\,\text{mV}$, and zero for the networks with a series capacitor (WE A and Cell 2). The mean ΔI over the scan was compared with this value, and the noise was taken as the standard deviation (Table 6.4).

Table 6.4: Ferricyanide DPV scans. Measured on its own drop, 0.8 V to 1.3 V

		Expected	Mean ΔI (nA)		Error (%)		Noise (nA)
Cell	**Network**	**(nA)**	**PS4**	**OpenStat**	**PS4**	**OpenStat**	**PS4 / OpenStat**
WE A	560 Ω + (33 nF ∥ diodes)	0	0.00	0.01	–	–	0.10 / 0.07
WE B	10 kΩ	2500.0	2504.4	2511.0	+0.18	+0.44	6.0 / 8.2
WE C	560 Ω + (33 nF ∥ 10 kΩ)	2367.4	2372.0	2380.5	+0.19	+0.55	5.4 / 6.1
Cell 1	93.8 kΩ	266.5	264.9	269.7	−0.62	+1.19	2.2 / 0.8
Cell 2	9.83 kΩ + 470 nF	0.05	0.11	0.21	–	–	0.38 / 0.09
Cell 3	556 Ω + (97.5 kΩ ∥ 1 µF)	255.0	254.9	259.7	−0.02	+1.88	1.0 / 1.4
Cell 4	556 Ω + (923 kΩ ∥ 1 µF)	32.4	33.1	32.4	+2.04	−0.08	0.7 / 1.2

Both instruments measured every network with a DC path to within 2.1 % of the expected value. The OpenStat read slightly high on all of them except Cell 4, by 0.4 % to 0.6 % on the 2.5 µA networks and 1.2 % to 1.9 % on the 260 nA networks. The blocking networks WE A and Cell 2 read below 0.25 nA on both instruments, so the charging current is rejected and there is no DC leakage path. The noise on these two networks, 0.07 nA to 0.09 nA, is the floor of the instrument itself; with the estimate of Section 2.1.4, its 3σ of about 0.25 nA corresponds to roughly 1 µg L^{-1} to 5 µg L^{-1} of efavirenz.

Figure 6.2: DPV scans of the dummy cell WE C (Left) and Cell 3 (Right) from 0.8 V to 1.3 V.



The next two tests replace the known load with a real electrode. From this point the true value is no longer known, so the PalmSens4 becomes the reference, and its own scan-to-scan variation becomes a variable.

6.4 Salt Solution Test

The fourth test replaces the passive circuit with a real electrode. A drop of 0.1 M KCl on the SPE has a solution resistance and a double-layer capacitance, but no redox reaction in

the scan window, which makes it a physical Randles cell with a very large charge-transfer resistance [19]. Figure 6.5b shows the setup used for the salt blank tests. It should give a smooth background without a peak. The OpenStat is judged against the PalmSens4 measuring the same solution with the same waveform parameters.

Five pairs of salt were compared, Salt 1 to 5 as seen in Figure 6.3, each consisting of one scan on each instrument on the same drop. Table 6.5 compares each pair quantitatively, using the mean current, the RMS difference and the correlation between the two voltammograms. The comparison runs from 0.40 V to −0.20 V, so that the ferricyanide peak of Section 6.5 lies in the window.

Figure 6.3: Salt solution measured by each potentiostat.

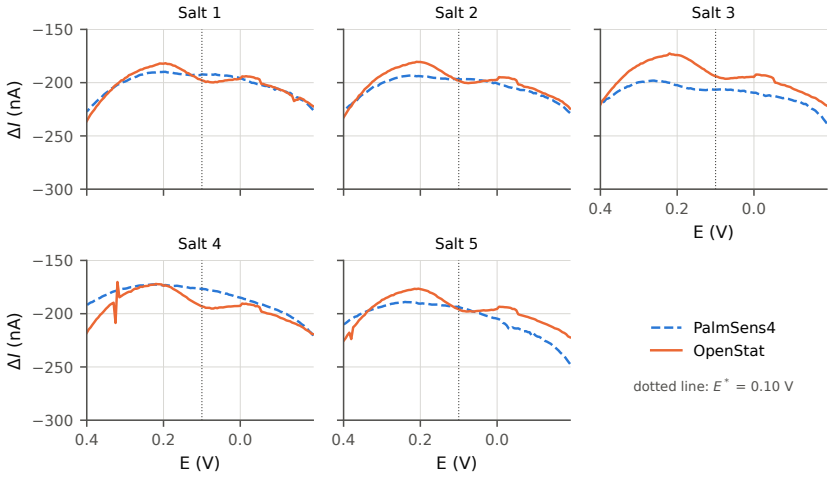


Table 6.5: Salt test. All scans on the OpenStat used the 80 kΩ range. r is the correlation coefficient between the PS4 and OpenStat voltammograms of each pair.

	Mean ΔI (nA)		Dev.	RMS	Noise (nA)	
Pair	**PS4**	**OpenStat**	**(%)**	**(nA)**	**PS4 / OpenStat**	r
Salt 1	−200.7	−200.0	−0.3	4.5	0.6 / 0.9	0.94
Salt 2	−204.0	−199.3	−2.3	6.6	0.6 / 0.8	0.93
Salt 3	−209.6	−194.0	−7.4	16.8	0.6 / 0.8	0.90
Salt 4	−184.7	−192.4	+4.1	10.2	0.5 / 3.1	0.86
Salt 5	−204.2	−196.4	−3.8	11.7	0.7 / 1.1	0.80

With no analyte present the current measured varies from −185 nA to −210 nA on the PalmSens4 and −192 nA to −200 nA on the OpenStat as shown in Figure 6.3 and Table 6.5. The mean deviation per measured pair ranges from −7.4 % to 4.1 % and the RMS difference

from 4.5 nA to 16.8 nA. This is larger than in the Randles test, but is of similar size as the variation of the cell itself. This is shown from scan to scan where the PalmSens4 measurement has a standard deviation of 10.4 nA, and the OpenStat 3.5 nA.

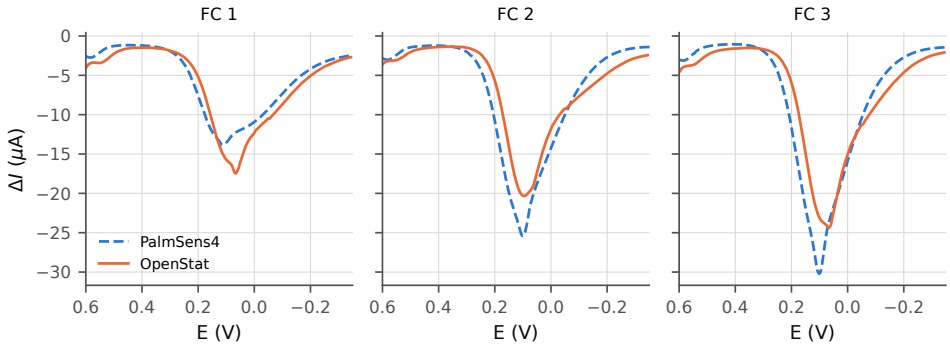
6.5 Ferricyanide Test

The last test adds a redox reaction. Ferricyanide gives a single, well-understood reduction peak, so it tests the potential axis and the current scale together [19].

$$[\mathrm{Fe(CN)_6}]^{3-} + e^- \rightleftharpoons [\mathrm{Fe(CN)_6}]^{4-}, \tag{6.2}$$

All OpenStat scans used the 1 kΩ range. Three pairs were compared, FC 1 to 3, each consisting of one scan on each instrument on a different drop as seen in Figure 6.4.

Figure 6.4: Ferricyanide DPV scans on the same drop, OpenStat run against PalmSens4 run.



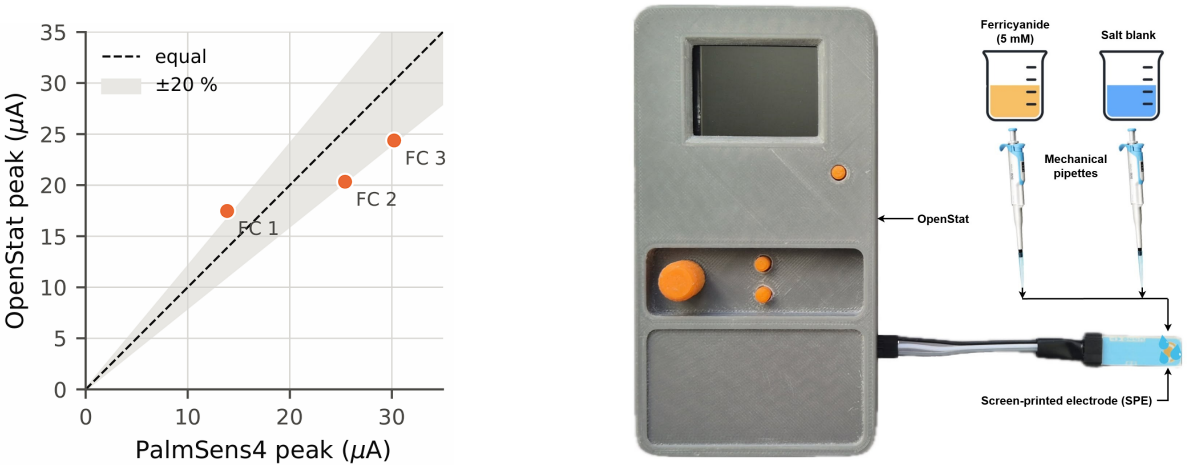
The OpenStat produces a peak similar in shape to the PalmSens4 shown in its correlation $r \geq 0.956$ for all three pairs. However, there are two visible differences: the peak heights differ by up to ±26 % and the OpenStat peak lies at a more negative potential (shifted to the right).

6.5.1 Peak Height

The peak heights of the three pairs are plotted in Figure 6.5a.

The OpenStat overestimates peak height by 26 % on FC 1 and underestimates it by approximately 20 % on FC 2 and 3, indicating bidirectional deviation rather than regular bias. This difference must be judged against the repeatability of each instrument. Consecutive PalmSens4 scans were taken to verify this, and they vary by 8 % to 52 %, while consecutive OpenStat scans vary by 3 % to 33 %. The mean difference between the instruments, 22 %, falls within the repeatability of the reference itself, so the test shows

Figure 6.5: The three ferricyanide pairs. (a) Peak height of the OpenStat against the PalmSens4. (b) OpenStat testing setup, either solution applied to the SPE by mechanical pipette.



(a) (b)

no instrument difference larger than the variation of the electrode, but cannot detect a smaller one. Because evaporation raises the peak over time, the later scans are expected to read higher.

6.5.2 Peak Potential

The potential offset was found by sliding the OpenStat curve of each pair along the potential axis until it matched the PalmSens4 curve most closely. The offset ranges from 20 mV to 27.5 mV, where 24 mV is the average. Shifting the OpenStat curve reduces the difference between the curves by up to a third. All the OpenStat ferricyanide scans used the 1 kΩ range, therefore the offset could potentially come from the specific range or its calibration. The calibration cannot cause it, as it only rescales the current. A range-dependent cause would act through a voltage drop proportional to the current, so the offset would grow with the peak; instead, it stays the same while the peak more than doubles from FC 1 to FC 3 (Figure 6.4). The OpenStat records each point at its base potential (Section 5.3.2), and the offset is close to half of the 50 mV pulse, which suggests that the instruments report the potential of a point at different positions within the step. Repeating a scan with a 25 mV pulse would test this, as the offset should then halve. Until the cause is confirmed, the offset must be corrected before peak potentials are compared with published values.

6.5.3 Sources of Deviation

The remaining differences come from the solution and the testing environment, which drift between scans. Because the two instruments never measure the same drop at the same

moment, this drift enters every comparison. Likely causes are:

- **Adsorption.** Gold is not inert in ferricyanide; when cyanide is released from the complex and adsorbs on the gold, the gold is slowly etched. This results in the active area and electron-transfer rate to change from scan to scan [63, 64]. This could explain the low and varying peak currents and the movement of the peak potential.
- **Light sensitivity.** Ferricyanide decomposes when exposed to light; it breaks down into ferrocyanide and releases cyanide [65]. During every scan the drop on the electrode is exposed to room light, which lowers its ferricyanide concentration and adds cyanide that attacks the gold.
- **Evaporation.** A small drop loses water within minutes, which raises its concentration and explains why repeat scans on one drop grow in peak height rather than shrink.

The comparison would improve by using a fresh electrode for each pair, shielding the drop from light, and starting each scan at a fixed time after the drop is placed.

6.6 Summary

Table 6.6 checks each objective of Section 1.3 against its target.

Table 6.6: Verification of the project objectives.

Objective	Target	Result	Status
1 Waveform	LSB ≤ 1 mV; pulse 25 mV $\pm$ 1 mV; period $\pm 1\,\%$	LSB 0.54 mV; pulse 25.17 mV; period $-0.7\,\%$; width $-3\,\%$ (about 1.5 % in current)	Met
2 Current	± 750 µA; sub-nA resolution; gain error $\leq 1\,\%$	0.23 nA per code; $\leq 0.44\,\%$ after calibration (reference $\pm 1\,\%$); overload at 720 µA in normal	Partly met
3 Agreement	Dummy cells; PalmSens4 within its repeatability	Dummy cells within 1.9 %; salt and ferricyanide within PalmSens4 repeatability; 24 mV offset	Partly met
4 Interfaces	Standalone and web browser operation	Both demonstrated (Figure 4.2, Section 5.3.2)	Met
5 Battery	At least 8 h	18 h to 24 h, estimated from measured current (Section 4.6.1)	Met (estimate)
6 Cost	Report and compare	R2 226 incl. VAT (Appendix D.12), against approximately R55 400 for the PalmSens4	Met

The bench tests confirm the electronics. The electrochemical comparisons were limited by the repeatability of the electrode rather than the instrument, and the 24 mV offset remains to be explained.

Chapter 7

Environmental and Sustainability Analysis

An instrument built for environmental monitoring should itself leave a small footprint, so this chapter considers the problem the OpenStat addresses and its impact over its life.

Antiretrovirals are not fully removed by South African wastewater treatment works and reach surface water at up to $102\,\mu g\,L^{-1}$ [24], far above the $ng\,L^{-1}$ levels at which efavirenz harms fish [66]. Monitoring is limited by the cost of laboratory analysis, which a low-cost portable instrument helps to lower. The components for the prototype cost R2 226 including VAT (Appendix D.12).

The largest savings were made in the hardware itself. The AD5941 integrates all the elements discussed in Section 2.2 into one millimetre-scale package [39], keeping the board small and the part count low. The SPEs, the only consumables, were cleaned with ethanol and reused throughout testing, which added to the variation seen in Section 6.5.3. Table 7.1 lists the remaining measures.

Table 7.1: Further life-cycle measures.

Stage	Design response
Energy	0.89 W measured load (Eq. (4.2)), about 1200 one-minute scans per charge
Obsolescence	Browser and on-device interfaces, so no companion app can become unsupported
End of life	Recycling under the EPR Regulations [67]

Charging the 18.5 W h cell through the linear charger draws about 29 W h from a grid emitting 0.960 kg CO_2e/kW h [68], roughly 28 g CO_2e per charge or 0.02 g per scan, so operating emissions are negligible.

The main remaining improvement is in the power chain: boosting the battery to 5 V and then dropping to 3.3 V in a linear regulator discards about a third of the stored energy, and a 3.3 V step-up/step-down regulator would extend the run time by roughly 50 % with the same cell. Safety, cost and societal considerations are discussed in Appendix B.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

This project set out to design, build and validate a low-cost, wireless potentiostat for the DPV detection of efavirenz and oseltamivir in water. Against the objectives of Section 1.3 (Table 6.6), the OpenStat applies the DPV waveform to within one DAC code (Objective 1); reads every current range to within 0.44 % after calibration but overloads above 720 µA instead of 750 µA when in normal mode (Objective 2, partly met). The OpenStat agreed with dummy cells to within 1.9 % and with the PalmSens4 within that instrument's own repeatability, apart from a 24 mV offset (Objective 3, partly met). The device can be controlled physically or from a web browser (Objective 4) for an estimated 18 h to 24 h per charge (Objective 5). It costs R2 226, against US$3 340 for the PalmSens4 (Objective 6). The electrochemical tests were limited by the repeatability of the electrode, and the target drugs were not measured, so the detection limit was not measured; the noise floor suggests roughly 1 µg L^{-1} to 5 µg L^{-1} for the electronics alone (Section 6.3).

8.2 Future Work

The 24 mV offset should be investigated first, by repeating a ferricyanide scan with a 25 mV pulse (Section 6.5.2) and on other current ranges. A ferricyanide concentration series on both instruments would then give the sensitivity and detection limit, and the noise floor should be measured with Wi-Fi on and off. In the firmware, starting each averaging window 20 ms before its edge would reduce the timing error (Section 5.4.4), raising C_f to 63 pF would improve the phase margin with small-capacitance electrodes (Section 3.2.2), and peak detection would let the device report peak height and potential. The hardware needs a grounded, shielded electrode cable (Section 4.4.3), the high-power mode of the potentiostat amplifier for the 200 Ω range (Section 5.5.1), and a charger with power-path control so that it can run while charging. The device should then be tested with efavirenz and oseltamivir, using preconcentration or a modified electrode; machine learning could then be used to identify and quantify compounds whose peaks overlap.

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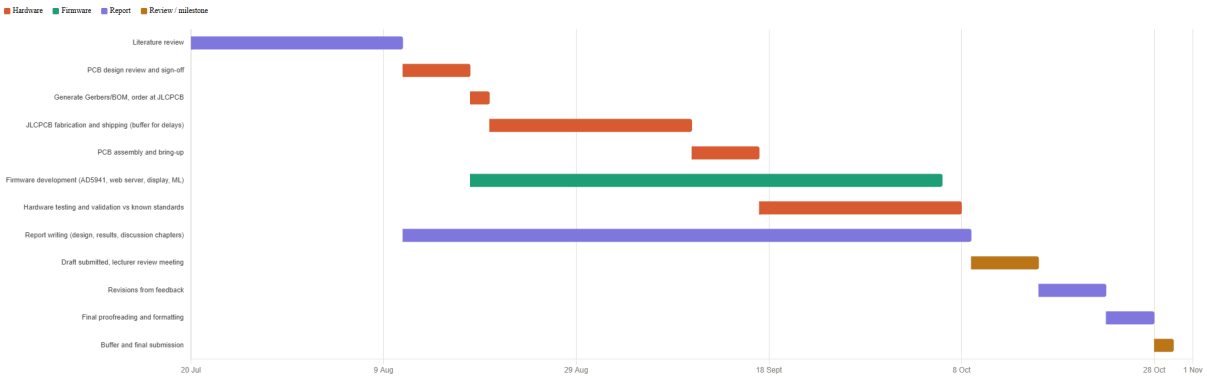
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Appendix A

Project planning schedule

Figure A.1: Gantt chart for the project planning schedule.



Appendix B

Graduate Attributes

B.1 Complex Engineering Problem

Designing a portable DPV potentiostat requires first-principles analysis, in-depth knowledge of electrochemistry, mixed-signal electronics and embedded software, and engagement with the current literature (Chapter 2). Nanoampere currents must be resolved while a Wi-Fi radio shares the supply, with the cell inside the feedback loops (Section 3.2.2). The requirements conflict in a public-health problem with no regulatory framework (Section 2.1.4), there is no obvious solution (Table 2.1), and no standard prescribes how such an instrument should be validated (Chapter 6). The work crosses into analytical chemistry (Section 4.1) and is a system problem spanning hardware, firmware and two user interfaces (Chapters 4 and 5).

Table B.1: Graduate attributes, with descriptions from ECSA document E-02-PE (revision 7), how each was achieved and the relevant sections of the report.

Graduate attribute	**How it was achieved**	**Relevant sections**
GA 1: Problem-solving *Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.*	The lack of a low-cost, standalone DPV instrument for antiretroviral residues in South African water was identified, formulated as an aim, scope and measurable requirements, and the literature was researched to locate the gap. The design was analysed from first principles, from the decay of the cell currents to the stability of the front-end, and unexpected results were reasoned out from the data. Sustainability and social context were considered, as these residues are not monitored in South Africa.	Sec. 1.1–1.4 Tab. 4.1, 4.2 Ch. 2 Sec. 2.1.3, 3.2.2 Sec. 6.2, 6.5.2 Ch. 7

Graduate attribute	How it was achieved	Relevant sections
GA 2: Application of scientific and engineering knowledge *Apply knowledge of mathematics, natural science, computing and engineering fundamentals, and an engineering specialisation to develop solutions to complex engineering problems.*	Mathematics was applied in the transfer functions and phase margin of the TIA loop, filter design, DAC quantisation and comparison statistics; electrochemistry to explain how DPV rejects capacitive current; computing in the firmware and averaging filter; and the instrumentation specialisation in the amplifier, grounding and power design. Formal modelling appears in the stability model, and engineering reasoning in tracing a common gain error to the shared conversion chain.	Sec. 3.2.2, App. D.13 Sec. 4.4.2, 5.3.1 Sec. 2.1.3, 6.5.3 Ch. 4, 5 Sec. 6.2
GA 3: Engineering design *Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs.*	The OpenStat was designed as a complete system to meet the requirements, comparing alternatives at each major decision. Its creative element is the dual interface: an onboard display for field use and a browser dashboard. For safety, no supply exceeds 5 V, the Li-Po cell is protected and ferricyanide is kept from acid and light. Components cost R 2 226, against R 55 400 for a commercial instrument.	Tab. 4.1, 4.2 Ch. 4, 5 Tab. 2.1 Sec. 4.6 App. D.12, Ch. 7
GA 4: Investigations, experiments and data analysis *Demonstrate competence to conduct investigations of complex engineering problems using research methods, including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.*	Antiretroviral residues are an emerging contaminant with no South African regulatory limit, and the investigation evaluated whether low-cost DPV can help address this. Five progressive tests each added one element of a real measurement, so that each deviation could be traced. Critical thinking was applied in judging agreement: correlation alone was not accepted, and the instruments were compared against the PalmSens4's own repeatability. Peak height proved to be electrode-limited, while the 24 mV offset remains unexplained.	Sec. 2.1.4 Ch. 6 Tab. 6.1 Sec. 6.5.1, 6.5.2 Sec. 6.5.3
GA 5: Use of engineering tools *Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.*	The tools created were the AD5941 driver (adapted from the Analog Devices library) with automatic range selection, a firmware averaging window, Randles dummy cells and the comparison scripts. Those selected and applied were KiCad, JLCPCB, PlatformIO, Git, the IPC-2221 calculator, the laboratory instruments and LaTeX. Limitations were recognised: instrument accuracy bounded the calibration to about ±1 %, and the integrated front-end limits the compliance voltage and current range.	Sec. 5.4.2, 5.5.2 Sec. 6.3, 6.5 Sec. 4.4.4 Sec. 6.2, 5.5.1

Graduate attribute	How it was achieved	Relevant sections
GA 6: Professional and technical communication *Demonstrate competence to communicate effectively and inclusively on complex engineering activities, both orally and in writing, with the engineering community and society at large, taking into account cultural, language, and learning differences.*	The report follows the module framework and the conventions of the discipline: requirements tables, flow diagrams, schematics, a bill of materials, numbered equations and IEEE referencing. For readers who are not electrochemists, chemical terms are explained where they first appear, and lists of symbols and acronyms are provided.	Whole report Front matter Tab. 6.6
GA 8: Individual and collaborative teamwork *Demonstrate competence to function effectively as an individual and as a member or leader in diverse and inclusive teams, and in multi-disciplinary, face-to-face, remote and distributed settings. (Individual aspects only.)*	Every stage, from the literature review to the PCB design, firmware, testing and report, was carried out individually, with weekly guidance from Dr T Ebrahim and help with placing the PCB components; AI use is declared in Appendix C. Time was managed with a schedule that included buffers. The work crossed into analytical chemistry, and unfavourable results and design oversights are reported openly.	App. A, C Sec. 4.1 Sec. 4.6.3 Sec. 6.6
GA 9: Independent learning ability *Demonstrate competence to engage in independent learning through well-developed learning skills.*	Electrochemistry, AD5941 configuration and mixed-signal design were learnt independently. Adaptability is shown by adopting new technologies, and engagement with emerging issues by studying antiretroviral residues as an emerging contaminant. Weekly feedback was worked in, and critical thinking was applied to the varying ferricyanide results. This sufficed for standard solutions, but not yet for the target drugs.	Sec. 2.1 Sec. 2.1.4 Sec. 3.2, 4.4 Sec. 6.5.3 Sec. 8.2
Sustainable development *Efficient resource use, environmental impact, whole-life cost, re-use of resources and net zero carbon.*	The project's purpose is itself sustainable: low-cost monitoring of water pollution. The integrated front-end keeps the part count low, operating emissions are negligible, and open-source firmware lowers the whole-life cost. Reused electrodes save material but are etched by ferricyanide.	Sec. 2.1.4 Ch. 7 Sec. 6.5.3
Critical thinking *How critical thinking played a part in the project.*	Both AD5941 amplifiers were weighed, agreement was judged against the reference's own repeatability, errors were traced from patterns in the data, and real water samples were left out until results can be independently confirmed.	Sec. 3.2.1 Sec. 6.5.1, 6.5.2 Sec. 1.4
Emerging technologies *How emerging technologies were evaluated or employed.*	Integrated front-ends were compared with discrete designs, adopted, and their limitations evaluated. Web-hosted instruments were compared with smartphone apps, and the web-hosted approach was extended with an onboard display.	Tab. 2.1 Sec. 5.5.1 Sec. 2.3.3

Appendix C

Use of Artificial Intelligence (AI) Tools

C.1 AI Work Declaration

Three AI tools were used in this project, each for a different purpose:

- **Claude (Anthropic)** was used for logical and complex tasks such as troubleshooting, code review, code generation, data-processing scripts and planning, and for editing, converting and checking report text.
- **GitHub Copilot**, within VS Code, was used for autocompletion of spelling and citations while writing.
- **Gemini (Google)** was used for restructuring language in the report.

I used AI as a tool to support my own work, not to replace it. Every output was reviewed, tested or checked by me before it was used.

In the project work, AI was used for the following:

- **Troubleshooting.** Diagnosing hardware and firmware problems during building and testing, by discussing symptoms and suggesting likely causes and checks, which I then carried out on the hardware.
- **Exploring solutions.** Used as an improved search engine, suggesting and comparing alternative approaches to design problems. I made the choice between them, for the reasons given in the report.
- **Literature search.** Suggesting relevant papers and search terms for Chapter 2. Every source was located, read and checked by me before it was cited.
- **Code review and improvement.** Restructuring weak or repetitive sections of the firmware, checking the code for loopholes and edge cases, and adding safety measures such as input limits, time-outs and overload handling (Section 5.5.2), as

well as helping me understand the AD5941 and how its drivers work.

- **Web dashboard code.** The HTML page served by the OpenStat's web server was generated by Claude from my description of the layout and the data it had to show. I integrated it into the firmware and tested it on the device (Chapter 5).
- **Data-comparison scripts.** Help with setting up Python scripts to process and compare the data measured on the OpenStat and the PalmSens4 (Chapter 6).
- **Planning.** Helping to plan the project schedule (Appendix A) and to break tasks into manageable steps, which improved time efficiency.
- **Calibration.** Help setting up the two-point calibration procedure and the calculation of the gain and offset constants for each RTIA range (Section 6.2, Appendix D.5). The measurements were taken by me, and the resulting constants were checked at the independent +0.25 V point before being compiled into the firmware.

In the documentation (this report), AI was used for the following:

- **Firmware chapter.** For Chapter 5 (Firmware Design), Claude was given my rough drafts of the flow diagrams and translated them into LaTeX (TikZ). I checked the diagrams against the code and the behaviour of the device, and corrected them where they did not match.
- **Derivations.** Claude was used to simplify and clarify the derivations in Chapter 3 and 4. I checked each step of the resulting derivations myself.
- **Drafting.** Text in Chapter 6 (Testing and Calibration), Chapter 7 (Environmental and Sustainability Analysis), Chapter 8 (Conclusion and Future Work) and Appendix B (Graduate Attributes) was improved with Claude, which flagged incomplete sentences and inconsistencies. I reviewed, corrected, edited and checked the content.
- **Graduate attributes.** Claude was used to help identify what each graduate attribute required. The evidence and discussion in Appendix B are drawn from my own work.
- **Conversion to LaTeX.** Chapter 4 (Hardware Design) was written by me as a rough draft in Word and converted into LaTeX with Claude, which also flagged incomplete sentences and inconsistencies for me to resolve.
- **Bill of materials.** Claude compiled the bill of materials for this report from the components I selected.
- **Language editing.** AI was used in Chapters 1 to 3 only to improve structure,

clarity and flow, mainly with Gemini.

- **LaTeX support.** Finding and fixing compilation errors, and checking that all figures, tables, equations and sections are correctly referenced.
- **Proofreading.** Checking spelling, with Copilot also completing citations while writing in VS Code.

All AI outputs were verified before use. Firmware changes were compiled, flashed and run on the OpenStat, and their behaviour was checked against the serial output. An oscilloscope was used to confirm that the code produced the correct DPV excitation waveform (Section 6.1). Not all AI-suggested code worked, some aspects failed and had to be debugged and corrected by me before they behaved as intended.

AI was not used to generate or alter any measured data. All results and conclusions in this report are based on measurements made during testing.

C.2 Own Work Declaration

In the project work, the following was done by me:

- **Project definition.** Defining the aim, scope and requirements of the project with the supervisor (Chapter 1, Tables 4.1 and 4.2).
- **Literature review.** Selecting the sources reviewed in Chapter 2.
- **Hardware design.** Selecting the components, designing the schematic and PCB layout in KiCad, ordering the board, and assembling and bringing up the prototype (Chapter 4).
- **Firmware.** Writing the firmware, designing the code structure, including current measurement and range selection, and making the design decisions behind them (Chapter 5). Except the content in C.1
- **User interface.** Designing the layout of the onboard screens and web dashboard, and how the user interacts with the device.
- **Testing and calibration.** Carrying out all measurements, including the waveform, resistor calibration, Randles circuit, salt and ferricyanide tests, preparing the test solutions and operating the PalmSens4 (Chapter 6).
- **Analysis and conclusions.** Interpreting the results and drawing the conclusions of Chapters 6 to 8.

In the documentation (this report), the following was done by me:

- **Writing.** Writing Chapters 1 to 3, including the original derivations, the rough draft of Chapter 4, the rough flow diagrams of Chapter 5, and the rough drafts and notes used for Chapters 6 to 8 and Appendix B.
- **Content.** Deciding what the report covers, and selecting the results, figures and tables presented in it.
- **Review.** Reviewing, correcting and approving the final text, figures, derivations and references of the whole report.

Appendix D

Additional Content

D.1 Capacitive Current Rejection in DPV

After a potential step, the total current at the electrode is the sum of the Faradaic and capacitive components,

$$I_{\text{total}}(t) = I_f(t) + I_c(t), \tag{D.1}$$

with I_c and I_f given by Equations (2.2) and (2.3). Expanding the two DPV samples into these components gives

$$\Delta I = I_{\text{end}} - I_{\text{before}} = (I_{f,\text{end}} + I_{c,\text{end}}) - (I_{f,\text{before}} + I_{c,\text{before}})\,. \tag{D.2}$$

I_{before} is sampled after the electrode has been held at the baseline potential for long enough that the previous capacitive transient has decayed, so $I_{c,\text{before}} \approx 0$. I_{end} is sampled at the end of a pulse that is long compared with $R_s C_{dl}$, so the transient caused by the pulse has also decayed and $I_{c,\text{end}} \approx 0$. Equation (D.2) therefore reduces to Equation (2.4), leaving only the Faradaic signal.

Because ΔI is the change in Faradaic current caused by a small potential change ΔE, it approximates the derivative of the current–potential curve:

$$\Delta I \approx \frac{\mathrm{d}I_f}{\mathrm{d}E}\,\Delta E. \tag{D.3}$$

For a reversible system governed by the Nernst equation, the current–potential curve is sigmoidal, and its derivative is a peak centred close to the half-wave potential $E_{1/2}$, offset from it by half the pulse amplitude [19, 69].

D.2 Display and SD Card Specifications

Table D.1: Key specifications of the DFRobot DFR0928 display [70].

Parameter	Value
Driver	ST7735S
Resolution	128×160 pixels
Interface	4-wire SPI
Supply current	33 mA at 3.3 V
Storage	SD card slot

D.3 Power Management Subsystem Specifications

Table D.2: Main specifications of the power subsystem components.

TP4056 [55]	**DW01A** [71]	**S9V11E2F5** [56]
Input supply voltage (V_{CC}): 4.0 V to 8.0 V	Overcharge detection: approx. 4.28 V to 4.30 V	Output voltage: fixed 5 V (+5 % / −3 %)
Preset charge voltage: 4.2 V with 1.5 % accuracy	Overdischarge detection: approx. 2.40 V	Input voltage range: 2 V to 16 V
Charge current: programmable up to 1000 mA	Overcurrent protection	Start-up voltage: 3 V; start-up current limit: 700 mA
Trickle charge threshold: 2.9 V	Low current consumption (1.5 µA)	Typical maximum output current: up to 1.7 A
Operating temperature range: −40 °C to 85 °C		Efficiency: typically 85 % to 95 %

D.4 Scan Flow and Timing Diagrams

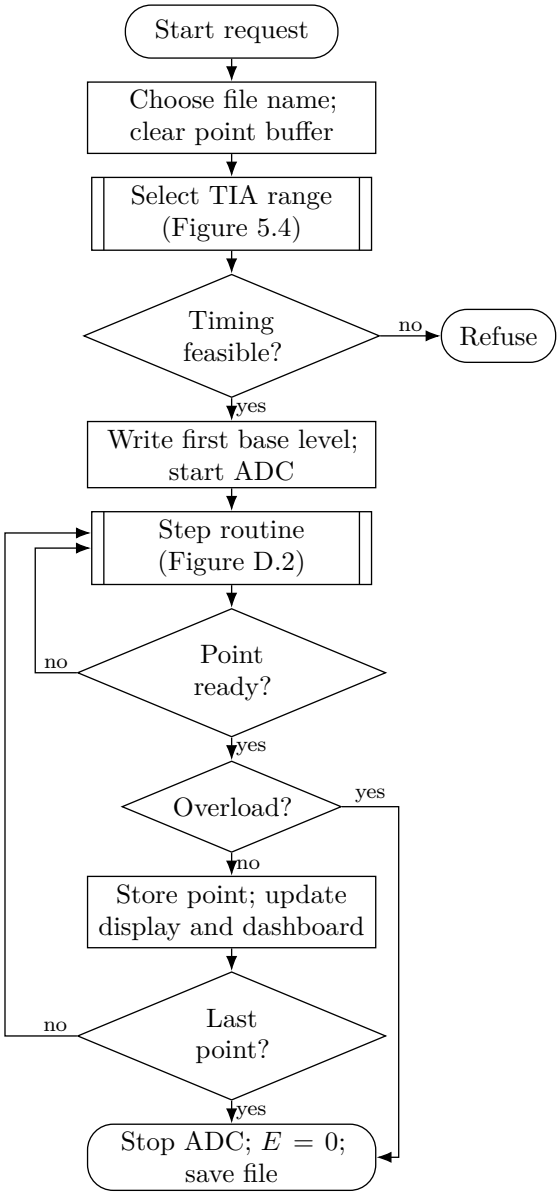


Figure D.1: Flow of one DPV scan.

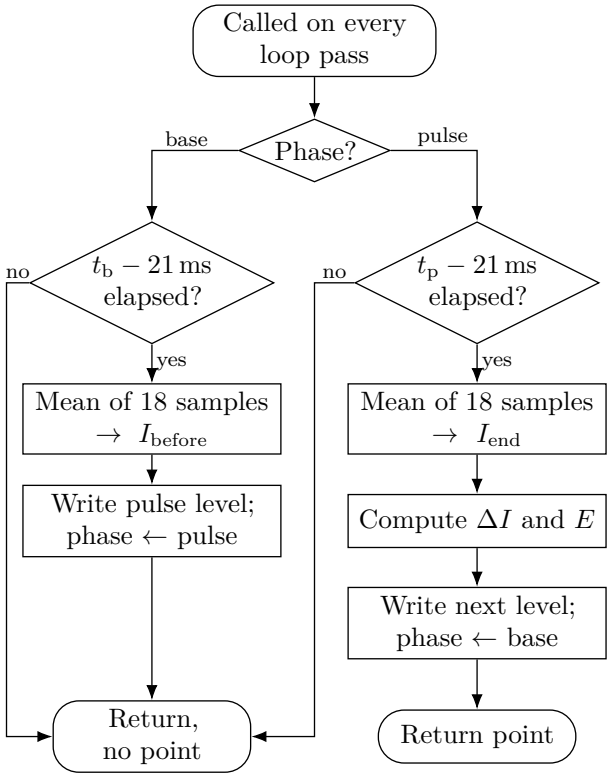


Figure D.2: Step routine. $t_b = T - t_p$ is the base dwell.

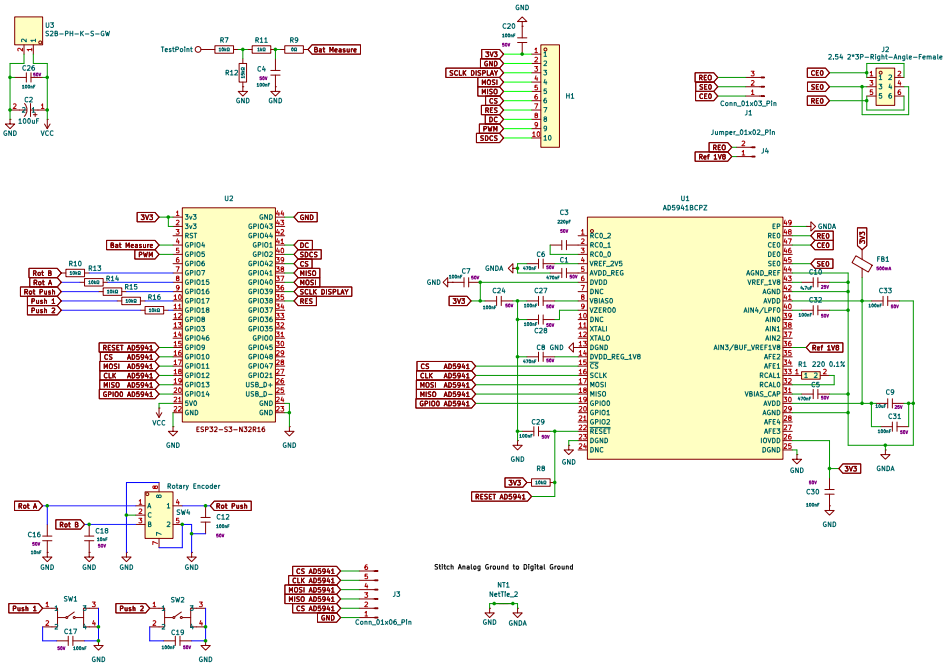
D.5 RTIA Calibration Constants and Resistors

Table D.3: Current ranges of the high-speed TIA, the calibration constants stored in the firmware and the resistor each range was calibrated against.

Code	R_{TIA}	Full scale	Resolution	G_k	O_k (nA)	Cal. resistor
0	200 Ω	±4.5 mA	187 nA	0.9013	+160.04	994 Ω
1	1 kΩ	±900 µA	37.3 nA	0.9141	+278.66	994 Ω
2	5 kΩ	±180 µA	7.47 nA	0.9089	−69.54	4.59 kΩ
3	10 kΩ	±90 µA	3.73 nA	0.8971	−81.57	9.93 kΩ
4	20 kΩ	±45 µA	1.87 nA	0.9018	−15.8	19.9 kΩ
5	40 kΩ	±22.5 µA	0.93 nA	0.9129	−9.159	38.5 kΩ
6	80 kΩ	±11.3 µA	0.47 nA	0.9158	−3.014	81.5 kΩ
7	160 kΩ	±5.6 µA	0.23 nA	0.9191	+1.005	146.7 kΩ

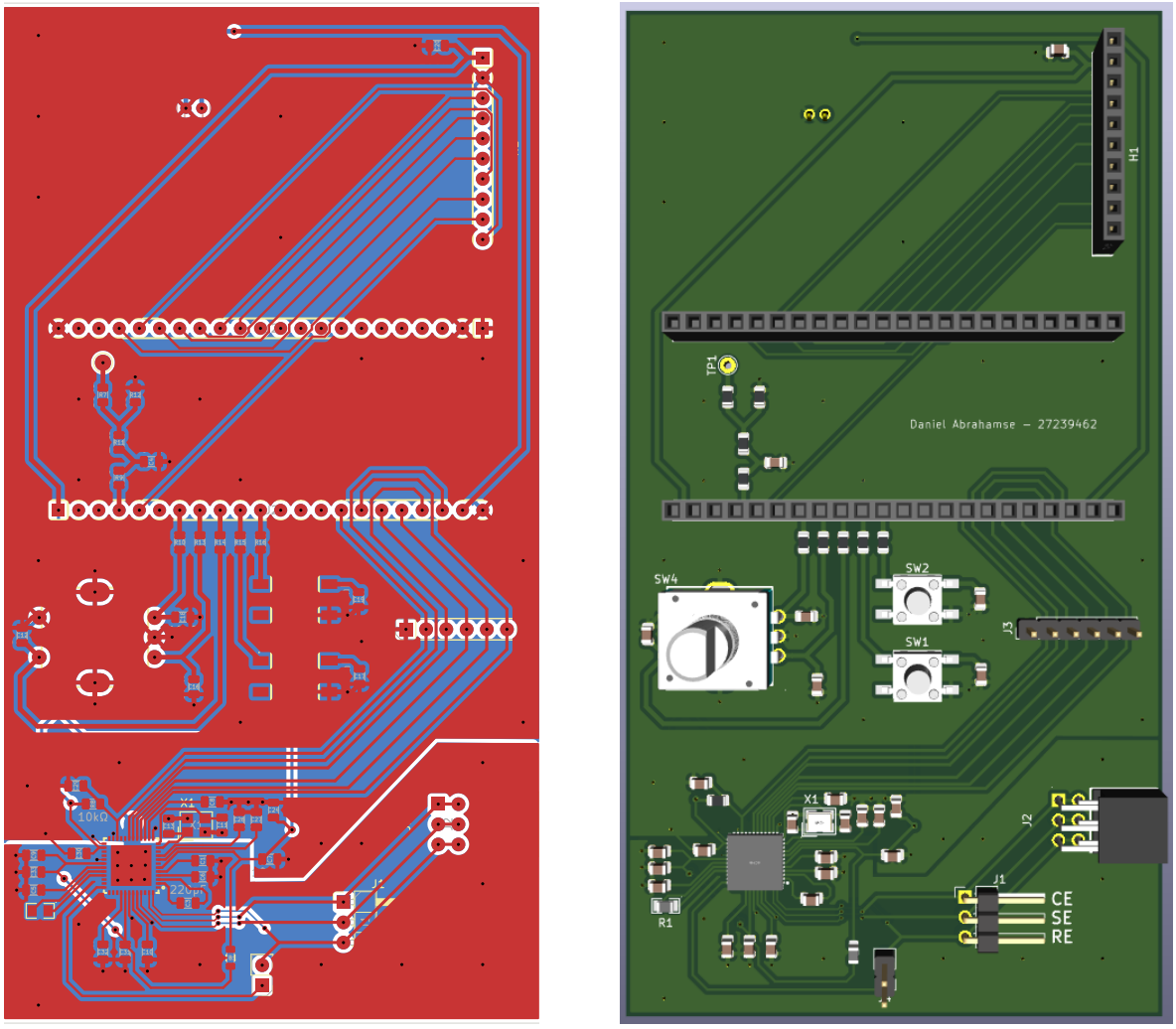
D.6 KiCad Schematic

Figure D.3: The KiCad schematic for the potentiostat.



D.7 KiCad PCB

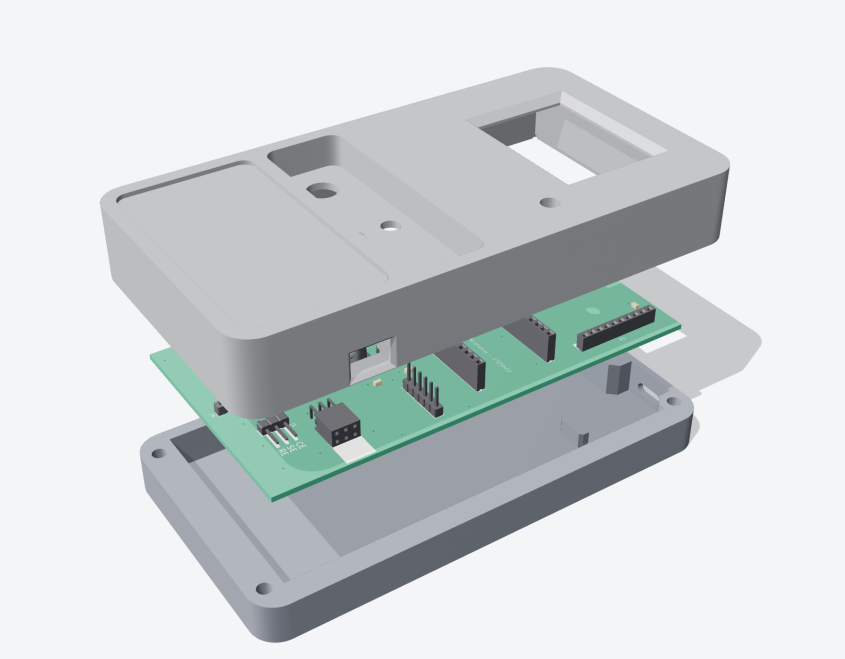
Figure D.4: The KiCad PCB for the potentiostat. (a) Copper layout. (b) 3D render showing the placed components.



(a) (b)

D.8 Case Render

Figure D.5: The case render for the potentiostat.



D.9 AD5941 Capacitor Selection

This appendix gives the full reasoning behind the capacitor values summarised in Table 4.3 of Section 4.4.1.

D.10 Supply Decoupling: 100 nF and 10 µF

Each supply pin has a local 100 nF ceramic capacitor, and AVDD has an additional 10 µF bulk capacitor. The MT-101 tutorial prescribes two tiers of capacitors: a large capacitor, typically 10 µF to 100 µF, within about 5 cm of the IC, and small ceramics, typically 0.01 µF to 0.1 µF, as close to the power pins as possible, optionally with a ferrite bead in series [43]. Analog Devices applies the same principle in its precision converter datasheets; for the AD7714, for example, a 10 µF capacitor is used in parallel with 0.1 µF on each analog supply [72].

Target impedance. The worst-case AVDD current step is the sum of all the subsystems of the AD5941 switching on at once [39, p. 14], which is approximately 10 mA.

The 1.82 V reference has an AC PSRR of 60 dB [39, p. 11], which corresponds to supply noise being attenuated by a factor of about 1000. A 10 mV supply disturbance therefore appears

as 10 µV on the reference, which is a factor of 5.5 below 1 LSB = $1.82\,\mathrm{V}/2^{15} = 55.5\,\mu\mathrm{V}$ [39, p. 7]. Taking 10 mV as a conservative limit on the supply disturbance, the target impedance is

$$Z_t = \frac{\Delta V}{\Delta I} = \frac{10\,\mathrm{mV}}{10\,\mathrm{mA}} = 1\,\Omega. \tag{D.4}$$

10 µF. The impedance of a capacitor falls as frequency rises, so a given capacitor holds the rail below Z_t only above a certain frequency. Setting the capacitor impedance equal to Z_t and solving for frequency gives that lower limit:

$$|Z_C| = \frac{1}{2\pi f C} = Z_t \quad \Rightarrow \quad f_{\min} = \frac{1}{2\pi C Z_t} = \frac{1}{2\pi\,(10\,\mu\mathrm{F})(1\,\Omega)} = 15.9\,\mathrm{kHz}. \tag{D.5}$$

Above 15.9 kHz the capacitor can supply the sudden current demands of the chip quickly enough that the AVDD voltage barely moves. Below 15.9 kHz the 3.3 V regulator holds the rail stable.

100 nF. Rapid switching inside the AD5941 draws short current bursts from the supply. The 10 µF capacitor cannot deliver bursts this fast. A small capacitor placed directly at each supply pin has far less inductance in its path and can deliver them, covering the frequencies from a few megahertz to beyond 100 MHz. A value of 100 nF is recommended for this role in standard decoupling guidance [43, 49, 73], and the calculations below show that it is adequate for this design.

Every capacitor resonates with its own equivalent series inductance (ESL) at its self-resonant frequency, above which it behaves as an inductor. With a typical ESL of about 1 nH for the 0805 package used on this board:

$$f_{\mathrm{SRF}} = \frac{1}{2\pi\sqrt{L_{\mathrm{ESL}}\,C}}: \qquad f_{\mathrm{SRF},\,10\,\mu\mathrm{F}} = 1.6\,\mathrm{MHz}, \qquad f_{\mathrm{SRF},\,100\,\mathrm{nF}} = 15.9\,\mathrm{MHz}. \tag{D.6}$$

The 10 µF capacitor is most effective up to around 1.6 MHz, which makes it the right part for the slower range, roughly 16 kHz to a few megahertz. The 100 nF capacitor keeps improving up to around 15.9 MHz. Above that its impedance is set by the ESL alone,

$$|Z| \approx 2\pi f L_{\mathrm{ESL}} = 2\pi\,(100\,\mathrm{MHz})(1\,\mathrm{nH}) = 0.63\,\Omega, \tag{D.7}$$

which is still below the 1 Ω target, so it covers the fast range adequately.

D.11 Internal Nodes: 470 nF and 4.7 µF

These capacitors connect to nodes inside the AD5941, specifically its voltage references and internal regulators. Their correct values depend on internal circuit details that Analog

Devices does not publish, so they cannot be derived independently and the datasheet values are used directly.

470 nF. The datasheet specifies 470 nF for three pins, namely `VREF_2V5`, `VBIAS_CAP` and `DVDD_REG_1V8` [39, pp. 11, 20–21], but does not explain how this value was chosen. Capacitors on such nodes are normally sized to suit the stability and noise requirements of the on-chip reference buffers and regulators [46], and those requirements cannot be assessed without internal design data. The datasheet value was therefore adopted as specified.

The datasheet gives no value for `AVDD_REG` [39, p. 20]. The 470 nF specified for the other internal regulator output, `DVDD_REG_1V8`, was used as the closest available guidance.

4.7 µF. The datasheet pin description recommends 4.7 µF on `VREF_1V82`, the 1.82 V reference of the ADC [39, p. 19], again without stating the reasoning. Its specification table, however, lists 0.47 µF as the test condition for the same pin [39, p. 11], so the datasheet is not internally consistent. The 4.7 µF recommendation was followed as the more conservative of the two. The ADC draws a small pulse of charge from this reference during every conversion, and a larger capacitor holds more charge, so the reference voltage dips less with each pulse.

D.12 Bill of Materials

Table D.4 lists the components bought for the prototype. Prices are the suppliers' listed prices in rand excluding VAT, for the quantity ordered; items sold in packs or with a minimum order are costed at the full pack. Shipping and import duties are excluded. Suppliers are abbreviated as MR (MicroRobotics), DK (DigiKey), CM (Communica), RS (RS Components) and JLC (JLCPCB). Items marked † are estimates. The PCB is costed from a JLCPCB online quote of US$30.89 for five boards at R16.59/US$ (25 October 2026).

Table D.4: Bill of materials for the OpenStat prototype.

Description	Part no.	Supplier	Qty	Unit (R)	Total (R)
Microcontroller					
ESP32-S3 development board, 32 MB flash, 16 MB PSRAM	N32R16V-M	MR	1	336.00	336.00
Display and storage					
1.8-inch 128 × 160 TFT display with microSD slot	DFR0928	MR	1	168.00	168.00
32 GB microSD card	MSC100-32GB	MR	1	148.00	148.00
Power					
Li-Po cell, 3.7 V, 5000 mA h	955565	MR	1	138.00	138.00
Li-Po charger module, 1 A, USB-C (TP4056)	PK4056	MR	1	16.00	16.00
Step-up/step-down regulator, 5 V	Pololu S9V11E2F5	MR	1	80.00	80.00
Electrolytic capacitor, 100 µF 25 V (pack of 10)	CAP-100UF-25V	MR	1	8.00	8.00
Analog front end					
Ferrite bead 0805, 600 Ω at 100 MHz	74279204	RS	10	3.98	39.84
Crystal, 16 MHz, ±10 ppm	X1E000021011912	RS	5	33.63	168.16
Thin-film resistor 0805, 220 Ω, 0.1 %	ERA6VEB2200V	RS	10	7.62	76.20
Connectors					
JST-PH 2-pin cable, 20 cm (pack of 4)	JST-PH-2P-20CM	MR	1	10.00	10.00
Edge connector, dual-row female, 6-way	7-5530843-7	DK	2	49.85	99.70
DIL socket, 6-way, right-angle	727060	CM	2	3.15	6.30
SIL header, 40-way	709401	CM	1	6.25	6.25
Female header, 40-way (pack of 10)	H40P-FEM-254	MR	1	18.00	18.00
SIL header, 20-way (pack of 10)	HM20P-254-BLK	MR	1	10.00	10.00
User input					
Rotary encoder EC11, vertical	EC11-VS-15	MR	2	10.00	20.00
Tactile switch, 6 mm × 6 mm SMD	DTSM63R	CM	4	3.70	14.80
Passives					
SMD electrolytic capacitor, 100 µF 6.3 V	100UF 6,3VES	CM	4	1.35	5.40

Table D.4 continued

Description	Part no.	Supplier	Qty	Unit (R)	Total (R)
Resistor 0805, 0 Ω (minimum order 100)	CHR0805 0R	CM	100	0.14	14.00
Capacitor 0805, 470 nF†	CHC0805	Kit	4	0.60	2.40
Capacitor 0805, 4.7 µF†	CHC0805	Kit	1	0.56	0.56
Capacitor 0805, 12 pF C0G†	CHC0805	Kit	2	0.35	0.70
Capacitor 0805, 100 nF†	CHC0805	Kit	15	0.94	14.10
Capacitor 0805, 10 nF†	CHC0805	Kit	2	0.45	0.90
Capacitor 0805, 220 pF C0G†	CHC0805	Kit	1	0.20	0.20
Capacitor 0805, 10 µF†	CHC0805	Kit	1	1.04	1.04
Resistor 0805, 10 kΩ, 1 %†	CHR0805	Kit	7	0.12	0.84
Resistor 0805, 1 kΩ, 1 %†	CHR0805	Kit	1	0.12	0.12
Resistor 0805, 15 kΩ, 1 %†	CHR0805	Kit	1	0.12	0.12
Prototyping					
Veroboard, 50 mm × 70 mm (pack of 4)	VERO5070	MR	1	20.00	20.00
PCB					
Two-layer FR-4 PCB, 150 mm × 50 mm (5 boards)†	JLCPCB	JLC	5	102.48	512.41
Subtotal (excl. VAT)					1 936.04
VAT (15 %)					290.41
Total (incl. VAT)					**2 226.45**

D.13 TIA Noise Gain with a Series Cell

At the inverting input of the TIA the cell is represented by the impedance $Z_c = R_\Sigma + 1/sC_{dl}$ to the solution node, which the control loop holds at a fixed potential, and the feedback network by $Z_f = R_{\mathrm{TIA}}/(1 + sR_{\mathrm{TIA}}C_f)$. Applying Kirchhoff's current law at the inverting node, with V_n the node voltage and I_{in} the faradaic current,

$$I_{\mathrm{in}} = \frac{V_n}{Z_c} + \frac{V_n - V_{\mathrm{out}}}{Z_f}, \qquad V_n = -\frac{V_{\mathrm{out}}}{A(s)}. \tag{D.8}$$

Eliminating V_n gives the closed-loop transimpedance

$$Z_T(s) \equiv \frac{V_{\mathrm{out}}}{I_{\mathrm{in}}} = -\frac{Z_f}{1 + NG(s)/A(s)}, \qquad NG(s) = 1 + \frac{Z_f}{Z_c}, \tag{D.9}$$

so the loop gain is $A(s)/NG(s)$ [28]. With $C_f = 0$,

$$NG(s) = 1 + \frac{sR_{\mathrm{TIA}}C_{dl}}{1 + sR_\Sigma C_{dl}} = \frac{1 + s\left(R_{\mathrm{TIA}} + R_\Sigma\right)C_{dl}}{1 + sR_\Sigma C_{dl}}, \tag{D.10}$$

which is Equation (3.1). Setting $R_\Sigma = 0$ recovers the photodiode case, $NG = 1 + sR_{\mathrm{TIA}}C_{dl}$, whose noise gain rises without limit.

With $A(s) = 2\pi\,\mathrm{GBW}/s$, the phase of the loop gain is

$$\angle L(jf) = -90^\circ - \arctan\frac{f}{f_z} + \arctan\frac{f}{f_p}. \tag{D.11}$$

For $f_c \gg f_z$ the second term is close to -90°, so $\phi_{\mathrm{PM}} = 180^\circ + \angle L(jf_c) \approx \arctan(f_c/f_p)$, which is Equation (3.2). Above f_p, $|NG| \approx 1 + R_{\mathrm{TIA}}/R_\Sigma$ and $|A| = \mathrm{GBW}/f$, so the crossover is at $f_c = \mathrm{GBW}\,R_\Sigma/(R_{\mathrm{TIA}} + R_\Sigma)$. A margin of 45° requires $f_c \geq f_p$:

$$\frac{\mathrm{GBW}\,R_\Sigma}{R_{\mathrm{TIA}} + R_\Sigma} \geq \frac{1}{2\pi R_\Sigma C_{dl}} \quad \Longrightarrow \quad R_\Sigma^2 \geq \frac{R_{\mathrm{TIA}} + R_\Sigma}{2\pi\,\mathrm{GBW}\,C_{dl}} \approx \frac{R_{\mathrm{TIA}}}{2\pi\,\mathrm{GBW}\,C_{dl}}, \tag{D.12}$$

since $R_\Sigma \ll R_{\mathrm{TIA}}$, which gives Equation (3.3).

With C_f fitted, Z_f falls above $1/(2\pi R_{\mathrm{TIA}}C_f)$, so the noise gain falls from its plateau at $-20\,\mathrm{dB}$ per decade and adds phase lead at the crossover.

D.14 Trace Widths

Equation (D.14):

$$A = \left(\frac{I}{k\, T_{\text{rise}}^{b}} \right)^{1/c} \tag{D.13}$$

$$W = \frac{A}{t \times 1.378} \tag{D.14}$$

The PCB is a two-layer board, so all traces are on external layers, for which $k = 0.048$, $b = 0.44$ and $c = 0.725$ [53]. A copper thickness of 1 oz/ft^2, an ambient temperature of 25 °C and a temperature rise of 10 °C were used. The required and chosen widths are given in Table D.5.

Table D.5: Required and chosen trace widths for a 10 °C temperature rise in 1 oz/ft^2 external copper.

Net	**Design current**	**Required width**	**Chosen width**	**Capacity of chosen width**
AD5941	10 mA	0.0005 mm (0.021 mil)	0.3 mm (11.8 mil)	1.00 A
Display	33 mA	0.0027 mm (0.107 mil)	0.3 mm (11.8 mil)	1.00 A
ESP32	1 A	0.3 mm (11.8 mil)	0.6 mm (23.6 mil)	1.65 A

D.15 Suitability of the Specified Analog Front End

Table D.6: The specified AD5941 compared to alternative analog front ends.

Option	1 mV **potential steps (Objective 1)**	**Sub-nA to** 750 µA **current (Objective 2)**	**Integration and effort**
Discrete op-amps, external DAC and ADC	Yes, depending on the DAC chosen	Yes, with suitable gain resistors	Many parts, noise and stability designed from scratch
TI LMP91000	No	Partly	Small, but needs an external ADC
ADuCM355	Yes	Yes	AFE and Cortex-M3 in one package
AD5941 (specified)	Yes: 0.54 mV LSB (Section 5.3.1)	Yes	Small; register-level driver required

D.16 Firmware, KiCad and Testing Scripts

https://github.com/Daniel-abr